

MoneyBrew

Final Report

Human-Computer Interaction 2025/26

Group T03G2

Ana Beatriz Sousa Pinto	201404488
Ana Margarida da Costa Ribeiro	202305264
Casemiro Melo Jorge de Medeiros	202301897
Tomás da Silva Morais	202304692

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Introduction

Throughout this semester, and in the context of the Human-Computer Interaction curricular unit, we developed a concept for a Finance-related mobile application, meticulously following the procedural phases presented in the theoretical and practical lectures. The process of designing our application concept, MoneyBrew, began in Phase I, with the definition of its general purpose, and the development and application of questionnaires to assess the interest of potential users in our concept. Then, in Phase II, we identified three core functionalities of the app, established a low-fidelity prototype that depicted the flow of execution of the three associated tasks, and carried out a heuristic evaluation experiment with our class colleagues. Finally, in Phase III, we took into consideration the feedback of heuristic testing, corrected the detected issues, defined our high-fidelity prototype, and performed user evaluations with individuals that fitted into the target audience of our concept. The present report explores in detail the MoneyBrew development process and how each stage contributed to the refining of its user interface, taking into account the various user experience feedback obtained in each of the three phases.

Part I: User and Task Analysis

Project's Idea Description

Financial uncertainty and instability have become pervasive issues affecting numerous individuals and households worldwide, driven by a rapidly changing global landscape characterized by rising prices, escalating inflation, and persistent economic volatility. However, many individuals perceive discussing, tracking, and managing their finances as a time-consuming, challenging, and often unpleasant task. To address this issue, a solution is being conceived, MoneyBrew.

MoneyBrew will be a mobile application designed to help worried, busy or unmotivated individuals and families manage their personal finances in a simple, fun and interactive way. With this product, users will be able to privately track their earnings and expenses, control their spendings, monitor financial progress, set and achieve savings goals, receive personalized financial tips, consult AI-powered assistance, dive into the world of investments, among many other useful and innovative features.

Our app will combine in one single place everything one needs to take control of their finances in an accessible, practical and, as valued by many, highly secure environment. It will be as easy as taking a sip of coffee.

Related Apps

Creating something completely new can be a little tricky, especially nowadays. Because of this, we searched for applications that act similarly or attempt to do what we envisioned. There are currently a lot of apps that are related to financial planning and investments, but we selected five that relate to our idea in different ways.

Goodbudget and *Monefy* are simple apps that solely allow users the possibility of creating a budget and keeping track of it. *MoneyWiz* is similar to the previous two, but offers a broader range of functionalities, which are related to the ones we have in mind for our app. It allows users to organize budgets per category, such as health, takeaway food, and more. It also enables establishing goals for a user's savings, summarizing all credit cards and bank accounts together and showing all recurring expenses on a calendar.

Magnifi, another related app, specializes in investment while relying a lot on artificial intelligence (AI) to carry out its features. For example, it uses AI to act as an investment advisor, analyzing the user's portfolio and providing insights about risk or potential adjustments. This AI agent can also find new opportunities for the user, so they can make faster movements and potentially more informed decisions.

The last app we examined was *Snoop*. This is the most complete financial planning tool we found. It can consolidate all of the user's bank accounts in one place, provides special offers and exclusive deals in order to save money, categorize spending, set monthly and/or individual budgets for each spending category. Budgets can be made manually or by the app based on each user's data. Additionally, the app can track the user's credit score (when applicable), support building their savings inside the app itself, and generate reports that alerts for any increase in expenses, refunds, and spending history. Lastly, *Snoop* can group all of a user's subscription payments, so they can visualize all of them in a single place.

Questionnaire Highlights

In order to better understand our target audience, we conducted a questionnaire, designed to investigate individuals' financial management habits, levels of financial literacy, and attitudes toward digital tools for expense tracking and investment decision-making. It aimed to identify behavioral patterns, challenges, and user expectations that could guide the design of an innovative financial management application.

The survey consisted of 23 questions divided into five thematic sections:

1. **Sociodemographic characterization**, including age, marital status, education, and professional situation;
2. **Current financial situation and typical earnings and spendings**;
3. **Investment practices and motivations**;
4. **Financial management habits**, such as budgeting and saving frequency;
5. **Opinions on a new financial management solution**.

The questionnaire included both multiple-choice and open-ended questions, with an estimated completion time of 5 to 10 minutes. It was developed and distributed online via Google Forms, allowing easy access and anonymous participation, and disseminated through social networks and e-mail. The survey remained open between October 2nd and October 5th, 2025, resulting in a total of 38 responses.

The main results of the questionnaire are presented below, with the complete questionnaire, as well as a summary of results and platform-generated graphical representations included in Annexes I and II, respectively.

Sociodemographic characterization

The sample was composed predominantly of young adults aged 18 to 25 years (78.9%), followed by 26 to 35 years (13.2%), 36 to 45 years (2.6%), and 56 to 65 years (5.3%).

Regarding educational background, 44.7% of respondents held a bachelor's degree, 34.2% completed secondary education (general academic tracks), while 7.9% completed professional secondary education. Smaller proportions reported holding a master's degree (5.3%), a Doctorate (5.3%), or basic education only (2.6%).

In terms of employment, 36.8% were students, 26.3% were working students, 23.7% were employed full-time, and the remaining identified as self-employed (5.3%), unemployed (5.3%), or public sector workers (2.6%).

Current financial situation

Household size was relatively balanced, with 36.8% living in households of four members, followed by 21.1% in three-person, and 15.8% in two-person households.

Household income levels varied substantially: 21.1% reported earning between 2000 and 2999€, while 18.4% earned between 1500 and 1999€, and an equal proportion of respondents were unsure of the income of their household. Lower income brackets (below 1000€) represented roughly 13% of the sample, whereas 10.5% reported earning between 3000 and 4999€, and, in equal percentage (5.3%) earning between 5000 and 7499€ or above 7500€.

Most respondents identified housing, utilities, and transportation as unavoidable monthly expenses, whereas entertainment subscriptions and leisure activities were the first to be considered expendable if financial constraints arose.

Investment practices

Regarding investment activity, 42.1% reported having some form of investment, while 47.4% did not invest at the time, but expressed interest in doing so in the future. Only 10.5% stated they had no intention of investing. This finding indicates a substantial openness to financial planning tools among the sample, particularly if supported by accessible and educational resources.

Financial management habits

A significant proportion of respondents demonstrated active engagement with their finances. 31.6% reported controlling their expenses daily, and 34.2% did so weekly, while 18.4% monitored their budget only for major expenses. Moreover, most people did follow through with their monthly budget, with 34.2% doing so every month of the past year, 13.2% in 8 to 11 of the months, and 15.8% managing to follow their budget between 4 and 7 months during the last year. However, savings behavior appeared more irregular: only 28.9% stated they managed to save every month, while 26.3% saved between four and seven months in the last year. Smaller portions of the participants saved between one and three months (15.8%), and 10.5% declared they never saved in the previous twelve months.

Regarding the method used to track their expenses, participants have used mostly notebooks, either digital (39.5%) or physical (28.9%), as well as Excel spreadsheets (23.7%). Additionally, the respondents identified the lack of time or motivation (43.2%), impulsive spendings (40.5%), last-minute expenses (38.7%) and lack of expense tracking (27%) as the main challenges that hinder their ability to further control their finances.

These results highlight that there is room for improvement in terms of financial management, particularly in regard to monthly savings, to which a new solution such as ours can potentially contribute.

Opinions on a new financial management application

Digital engagement remains limited: 55.3% of respondents have never used a financial management application, while 18.4% have used one in the past but no longer do so. Only 26.3% of participants currently use such tools with varying frequency – 10.5% weekly, 10.5% monthly, and 5.3% daily. These numbers suggest that while awareness of financial management apps exists, sustained usage and long-term adoption remain low, likely due to usability or motivational barriers.

As for a potential new solution that could help users tackle their financial management challenges, the respondents consider simplicity, and security and privacy as their main interests (76.3% each), followed by an intuitive and appealing visual interface (57.9%), and integration with external services (34.2%). In terms of specific innovations, detailed analysis of consuming habits (67.6%), direct expense management (51.4%), financial literacy learning (40.5%), and AI chatbot help (24.3%) were highlighted by the survey respondents. As such, all these points of interest are being taken into consideration when designing the features and user interface of our app.

PACT Analysis

Having completed the collection and organization of the results, they were further interpreted to identify key trends and general patterns using a PACT analysis. This aimed to clarify the main characteristics of the surveyed **People**, the general **Activities** they currently engage in that are related to financial management, the **Context** in which those activities are carried out, and the **Technologies** they use to do so.

People

Our sample can generally be classified as consisting of young individuals, most predominantly between the ages of 18 and 35. The vast majority of them are single and hold either a high school diploma or a bachelor's degree. They are primarily students, either concurrently working or not, or non-self-employed workers. Their households typically comprise 3 to 5 people, including at least one family member who is financially dependent on the others. The average monthly household income typically ranges between 1500 and 2999€.

Activities

Regarding their day-to-day financial management activities, the respondents to our questionnaire reported monitoring their spending on a daily or weekly basis and typically create a monthly spending budget that they have followed through most months of the past year. However, these efforts generally did not translate into a consistent savings pattern. There is a notable interest in investing, as most participants already invest or, if they do not, wish to do so in the future. Moreover, although people reported a diverse range of recurrent monthly expenses they consider non-discretionary, entertainment-related spendings, such as entertainment subscriptions, physical exercise activities or TV channel packages, were standout areas where participants could cut back, if necessary.

Context

Regarding the context surrounding the reported financial-related activities, respondents indicated that a lack of time or motivation, inability to control impulsive spendings, failure to track expenses, and a hard time saving or predicting regular expenses are the reasons why they struggle to manage their finances. Additionally, despite their interest in financial investments, they admit to not having extensive knowledge in this field.

Technologies

As for the technologies used in relation to the mentioned activities, physical or digital notebooks and Excel spreadsheets are the most common methods people have used in the past for tracking expenses. However, a considerable percentage of surveyed people have never tracked their expenses at all. Most participants do not currently and have never used a financial management app, but recognize expense tracking, budgeting, and consumption pattern analysis as relevant features in the currently available options.

Personas

After gaining a deeper understanding of the individuals, we created two different personas. These personas represent the main target groups that our application aims to serve, summarizing their goals, motivations, and challenges in a concise way. Visual representations of the two personas we developed are available in Annex III.

Persona 1: Maria Alves

Background and lifestyle

Maria Alves is a 22-year-old student from Porto, Portugal with a bachelor's degree in Psychology. She is currently single and lives in a middle-class household with her parents and younger brother. As a master's student in Psychology at the Faculty of Psychology and Education Sciences of the University of Porto, Maria envisions herself working in an international and successful company, managing human resources. In addition to her studies, Maria works part-time as a waitress at a local coffee shop to help pay her tuition fees, support her parents on their family's expenses, and save a little money for herself. In her free time, she enjoys going to the gym, watching TV shows and spending time with her friends. Currently, Maria's main goal is to move out of her parents' house after finishing her studies; however rising rent prices and the increasing cost of living make her anxious about achieving financial independence.

Goals and motivations

Maria's main financial goals include closely monitoring her spending and earnings, tracking her savings in order to reach her financial goals, starting to invest to protect her money from inflation, learning more about personal finances, and developing overall good financial habits.

Frustrations and pain points

Maria faces several challenges when it comes to managing her finances, particularly a lack of consistent registration of personal spending and earnings, mainly due to low motivation, and difficulties in planning her financial path and defining a savings plan because of unpredictable expenses and the rising cost of living. Additionally, she has limited knowledge about investments, and no clear guidance or support to help her get started.

Technology preferences

In her daily life, Maria prefers using mobile devices and values apps that integrate with external services and offer cloud syncing. She also favors cross-platform and multi-device applications. Regarding potential financial management apps, she values a visually appealing and intuitive interface, prefers to be guided when using new software or unfamiliar features, and worries about privacy and data security.

Persona 2: Bernardo Martins

Background and lifestyle

Bernardo Martins is 41 years old and lives in Setúbal, Portugal with his wife and teenage son. Although he holds a bachelor's degree in Business Administration, Bernardo now works as a real estate agent at Prime Properties, earning a commission-based salary that contributes to his family's medium to upper-medium class status. As a real estate agent, he is known for his strong sales skills, but is consistently challenged by the ups and downs of commission-based income. Bernardo keeps a close eye on the market and world events, and likes to unwind with true crime TV series and podcasts. He tries to keep track of commissions and expenses on paper or Excel, but his financial choices, often shaped by friends' advice, are usually more reactive than strategic.

Goals and motivations

Financially, Bernardo aims to better monitor his commission earnings from multiple property sales and find a way to create monthly financial reports for tax purposes. He also wants to prepare for his son's university education and for his and his wife's retirement. To maximize his earnings, another key goal of his is to gain a better understanding of investment opportunities.

Frustrations and pain points

Bernardo has identified several factors that hinder his financial control, including unpredictable earnings that make it difficult to plan long-term budgets, having to manually track his finances using spreadsheets, which is time-consuming and error-prone, and also challenges in reconciling payments from different clients. He also has limited investment knowledge, relying on intuition or advice from his friends. Addressing these issues would significantly help Bernardo achieve his financial goals.

Technology preferences

When considering a potential app that helps him better manage his finances, Bernardo expressed interest in a tool that allows him to quickly and easily access his finances, offers various mobile notifications, integrates with banking apps and services, allows the generation of exportable reports, and provides educational tips about saving and investing. He also values visual data representation over text and prefers a simple and intuitive interface with minimal clicks.

Activity Scenarios

Having developed these two personas, we took the process a step further by creating hypothetical scenarios illustrating how each persona might interact with the app in real-life situations. These scenarios help us better understand how the user would engage with the application and how it could be seamlessly integrated into the everyday lives of our target audience.

Activity Scenario 1: Maria Alves

Maria is on a break between college lectures and receives a notification from the MoneyBrew app, telling her that a product that she generally buys on a weekly basis, Monster

energy drinks, at the Continente supermarket is now at a 20% discount. The app suggests that she buys 12 cans, which it determined should last her a month, given her past buying habits. Doing this, Maria would be able to save 4.20€. She sets up a notification for 6 pm, time at which she predicts she will go make her weekly purchase at that supermarket, so that she doesn't forget to follow the app's recommendation. At that time, she proceeds to buy the recommended products, which the app confirms by accessing her supermarket receipt, and removes the Monster energy drinks from the shopping lists for the following three weeks.

Activity Scenario 2: Bernardo Martins

Bernardo comes home after a long day of work and, after having dinner with his wife and son, begins winding down for the day. Then, he receives a notification from the MoneyBrew app, suggesting to him a new true crime Netflix series that has just arrived in their repertoire. The app has analyzed his Netflix consumption patterns and, two months ago, recommended him to cancel his subscription, because it believed no new series would capture his interest during that time. Excited by the app's summary of the new series, Bernardo decides to repurchase a Netflix subscription. After a hard day's work, he feels this is the perfect way to relax before bed. Thanks to the previous two-month cancellation, he saved about 20€.

Functionalities

For the MoneyBrew application to fulfill its objective of providing a simple, intuitive, fast, and secure way for busy or unmotivated individuals to manage their personal or familiar finances, a set of functionalities that contribute to this goal must be developed and implemented. With this fact in mind, the following features were conceived to be posteriorly implemented in the system, ensuring they are time-efficient, fully automated whenever applicable, and as most user friendly and intuitive as possible.

Bank account integration

As this feature represents one of the most valued aspects for potential users, MoneyBrew will communicate with the major bank applications, in order to allow users to manage their bank accounts, perform, validate, restrict, or even block specific transactions, all directly through MoneyBrew. Furthermore, all transactions made through a user's bank account will be automatically registered in the app, eliminating the need for direct intervention of the user for that particular financial data of theirs to be present in the system. The privacy of the users' data and the security of the system during these connections between the application and banking services will be fully ensured.

Financial history and analysis

Our system will enable the user to access detailed records and visual summaries showcasing their financial activities and illustrating the evolution of their financial status over a defined period. Automatically generated analytical reports, including key financial indicators in an intuitive and easily interpretable format, will be provided to the user, offering a comprehensive overview of their financial situation. Furthermore, the app will deliver personalized recommendations and insights aimed at improving the user's financial situation and future performance. In addition, all income and expenditure entries will be systematically

listed, explicitly identified, and adequately categorized, being provided for the user to consult, search and filter their financial data at any moment, refining a feature that is already available in similar applications, and that potential users identified as one they highly value.

Savings plan creator and tracker

Matching what is available in similar systems and is highly appreciated by our sample, MoneyBrew will enable users to define saving goals and monitor their progress effectively. Users will be able to configure specific saving parameters, such as the target amount and the desired monthly contribution, and see their progress, which will be represented via a progress bar in the app. Additionally, the system will be able to automatically restrict transactions that may compromise these goals, force the user to explicitly confirm those transactions, or even adapt the plan after a purchase that negatively impacts the user's budget, alerting users when their spending habits are negatively impacting their progress. The application will also provide personalized tips to support users in achieving their objectives and generate notifications upon reaching a goal or milestones, such as attaining 50% of the target amount or completing the saving goal.

Expense management

One of the most notable innovative features of MoneyBrew, which the respondents to our questionnaire identified as being one they would mostly like to see being implemented, will be the possibility of viewing, adding and removing expenses, directly through the app. As already mentioned, expenses will be systematically displayed and categorized in the system, and recurring ones, such as monthly subscriptions to streaming services or gym memberships, can be managed more deeply, by allowing users to add or remove them and stop any transaction that relates to it directly from the app. In some cases, this process can be entirely automated. For example, if the user specifies that they intend to watch a specific TV show, the app will automatically pay the subscription to a streaming service that adds it to its catalogue. Similarly, if the user has not utilized a particular streaming service for an extended period, the application may, based on content preferences and upcoming catalog updates, automatically cancel the subscription or recommend its cancellation. In the same way, the system may suggest or reinstate a subscription if new content is added that aligns with the user's interests. The security and privacy of the users' data during the connections between MoneyBrew and other external services will be completely ensured.

Smart notifications

To inform users about specific events or deliver relevant information, such as recommendations, price alerts, or milestone achievement, the app will issue personalized notifications, which will be easy and quick to read. The frequency and scope of these notifications can be configured by the user, ensuring a non-intrusive experience tailored to their individual preferences.

Personalized financial tips

Based on the current saving objectives, long-term goals, historical financial trends or the overall financial condition of the user, the application will provide personalized recommendations and insights designed to assist users in improving their financial well-being.

Localization-based recommendations

MoneyBrew will use the localization of the mobile device, when enabled, to provide recommendations to the user based on its current location. For example, if the app detects that the user is inside a particular supermarket, it will give the user information such as a list of the products with the best price-quality ratio or the best promotions. With this innovative feature, users who are typically in a hurry or do not have the motivation to plan ahead and write a shopping list beforehand will have the possibility to visualize a suggested shopping list right as they arrive at the supermarket, based on past purchases and the current promotions. They will then be able to actively contribute to their financial goals easily and without needing to directly access the app and look for this information.

AI assistant

MoneyBrew will feature *Brewy*, a generative AI chatbot that users can interact with at any moment. Brewy will be able to clarify the user's doubts, provide information, give personalized recommendations, or simply make the user engage with a supportive companion throughout their journey toward financial well-being. All data accessed and provided to Brewy will be processed locally, in order to avoid potential leaks of private or sensible user information.

Financial Literacy Hub

To fulfill the need for improved financial literacy identified in our user analysis, MoneyBrew will include a section where users can browse and search through a variety of educational content about personal finances, including articles, videos, graphs, among others, in order to strengthen their financial knowledge. This hub will go through various topics such as taxation, financial planning models, and investments. Quizzes and simulations about each subject will also be available, allowing the user to reinforce their newly acquired knowledge.

Investment planning and management

The great majority of potential users of our app invest or, at least, plan to start investing. So, in order to satisfy their needs, MoneyBrew will enable users to create a guided investment plan by specifying investment objectives (short-term or long-term returns) and preferences regarding investment types (stocks, ETFs, cryptocurrencies), among other parameters. Throughout both the planning and execution phases, the platform will provide comprehensive information to support informed decision-making by the user and ensure they remain confident and well-guided. This will include real-time price evolution of selected investments, market trends, company information, dividend yields, among other metrics. All owned investments will be clearly displayed, accompanied by visualizations illustrating overall price trends. Finally, users will be able to execute transactions, buying or selling investment products, at any time during market hours.

Smart price alerts

MoneyBrew will include the option for a user to receive a notification whenever a specified product or asset reaches a certain price. Additionally, the user will be able to configure the system to automatically execute transactions, such as purchasing the product or buying/selling the asset, once the target price is reached, without the need of direct user intervention after the configuration.

Part II: Lo-fi Prototype and Heuristic Evaluation Report

Project Abridged Description

MoneyBrew is being designed with the purpose of being the all-in-one central for one's financial life, thereby helping users take control of their finances on various levels. As such, this app will be able to accomplish a multitude of tasks, from tracking incoming and outgoing monetary transactions, to guiding the users into achieving their goals, while also providing a platform for active learning. However, for the purpose of the current phase, we focused on three specific tasks for which we developed a functional lo-fi prototype. These selected tasks reflect increasingly complexity in terms of number of steps to conclude the task, and also highlight three innovative functionalities MoneyBrew will support: the Brewy AI chatbot (task 1), the smart and automatic notifications for expense management (task 2), and the financial literacy hub (task 3). In particular, the three tasks tackled at the current phase were as follows:

1. **Task 1:** ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model is;
2. **Task 2:** remove an inactive Spotify subscription after receiving an automatic notification informing the user that they haven't listened to music for over 10 days;
3. **Task 3:** access a specific learning page (with video and article) about Work Contracts. After reading and watching, take a small quiz about the learned content.

Functionality 1: Brewy AI chatbot

For this functionality, the simplest of the three, the goal was to implement and evaluate the capability of the users to ask the AI chatbot any finance-related question. For our concrete implementation, the specific task was to ask Brewy to explain what the 50-30-20 financial model is.

For all three tasks, we defined usability requirements that the app should support. For this first task, and with respect to efficacy, the goal is that Brewy provides a detailed explanation of the 50-30-20 budgeting method, explaining what corresponds to each of the percentages that make up its designation.

To better qualify this parameter, performance indicators require that the chatbot outputs the correct definition of the model, and displays and clarifies what each of the percentages (50, 30 and 20%) corresponds to ("Needs", "Wants", "Savings"). The target is that, on average, no more than 2 prompts are required for the user to receive the expected answer, which should be accurate and representative of the requested information 95% of the times.

As for the efficiency metrics, the user should obtain the required output almost instantly. In particular, the task should be completed within 20 seconds, reaching the chatbot should not take more than 3 taps from anywhere within the app itself, and output presentation should require a single tap after writing the prompt.

Satisfaction, in this case, is represented by an overall positive reaction by the user, defined by a score of at least 4 out of 5 on a satisfaction scale, positive feedback in terms of ease of use and learning, trust in the provided information, significant knowledge attained, as well as its subsequent implementation on their financial life, and the minimal occurrence of user reports regarding this feature.

Functionality 2: Smart and automatic notifications for expense management

In terms of the actions upon receiving a smart notification, the example task we defined was for the user to remove an inactive Spotify subscription (music streaming service) after receiving such a notification informing them that they had not listened to music on that platform for over 10 days.

The efficacy of this task should be such that the user must be able to successfully remove the inactive subscription after receiving the automated suggestion. Particularly, and after the user carries out the task, the subscription should be correctly removed from the list of active recurring expenses, and the total monthly expenses should automatically update to reflect this change. In general, at least 95% of the users must complete the task without errors, and no incorrect deletions or inconsistent expense records should occur.

To evaluate the efficiency, we established concrete performance indicators: the time it takes to fully complete the task should be under 2 minutes, and the number of user actions required to do it should be no more than 5. Overall, the task should be completed quickly and effortlessly. In this particular case, the automated detection of the inactive subscription and consequent suggestion should reduce the need for manual expense review by the user by at least 40%, making it a considerably beneficial automation.

In terms of satisfaction, the user's subjective reaction to using this procedure should be assessed positively, with an average score of 4 to 5 out of 5 in a post-task satisfaction questionnaire, and at least 85% of users finding the task easy and simple to understand and execute. Furthermore, positive feedback about the automation, with users feeling secure and in control when allowing the system to manage external financial connectors is of particular importance.

Functionality 3: Financial literacy hub

The third and most complex task of the selected group details that users should be able to access certain educational content in the form of a video and an article, specifically about Work Contracts, and, after analyzing it, be able to take a simple quiz about that content. This is an example of the capabilities of the financial literacy hub, whereby users can further expand their knowledge and better understand the intricacies of financial management.

Regarding its efficacy, the task should be accomplished with the user learning the specific content in question and applying that freshly learned knowledge into their own personal finances. Concretely, the task is to be considered to have a positive efficacy if the quiz completed after the lesson is done so with a positive score, and if the user is able to explain how they can implement that knowledge in their daily life.

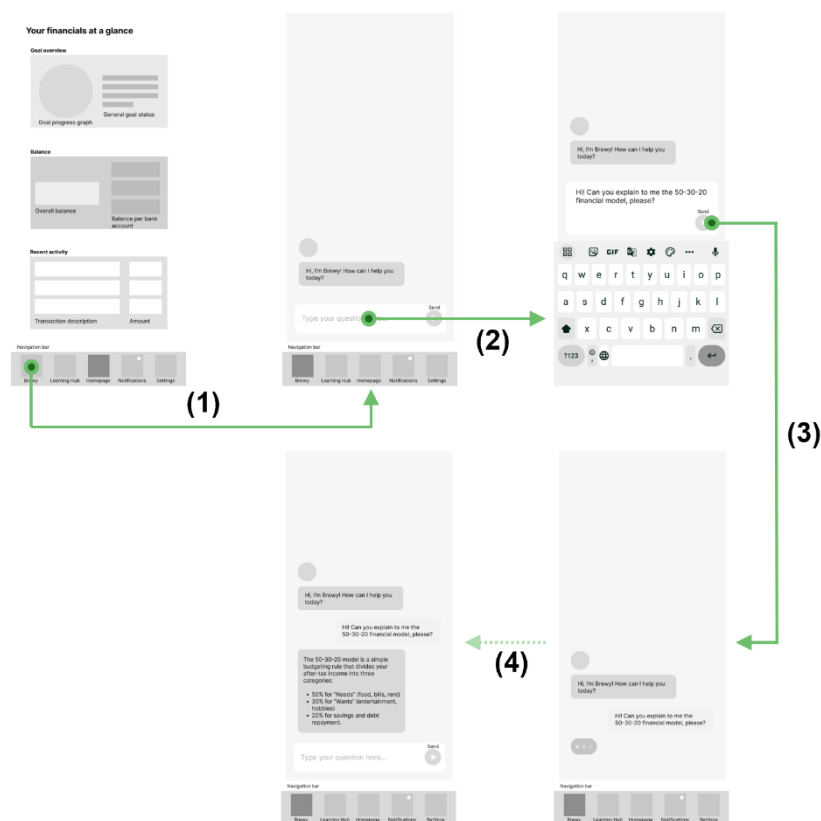
When it comes to its efficiency, the performance indicators of the task that we defined consist on a completion time, on average, between 5 and 10 minutes, and a number of required user interactions no more than 12.

As for satisfaction evaluation in the context of this task, it is expected that users offer positive feedback and opinions on the provided materials and the overall task as a whole, with an average score of about 4 out of 5, and limited number of reports regarding general problems and content-related issues, as the presented information should always be verified and updated when needed.

Prototype's Wireflow

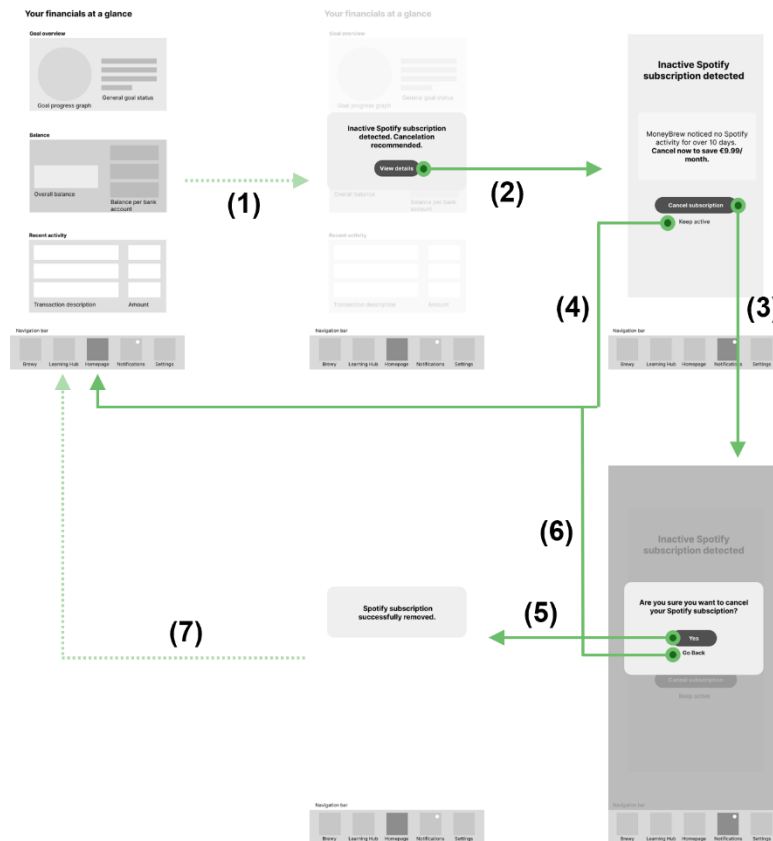
After careful consideration of the tasks on which we would focus for this particular phase, we designed the first draft of a functional and low-fidelity prototype that elucidates the expected pathway users are expected to take to accomplish the aforementioned tasks. As such, the present chapter details the wireflow of each individual task. We opted to start each wireflow on the homepage screen, as a central point of the app's navigation; however, execution could begin from anywhere on the app for tasks 1 and 3, as the relevant screens are easily accessible via a consistently-present navigation bar, and from clicking the notification for task 2, independently of where the user was within the app at the time they received that alert.

Task 1: Ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model is



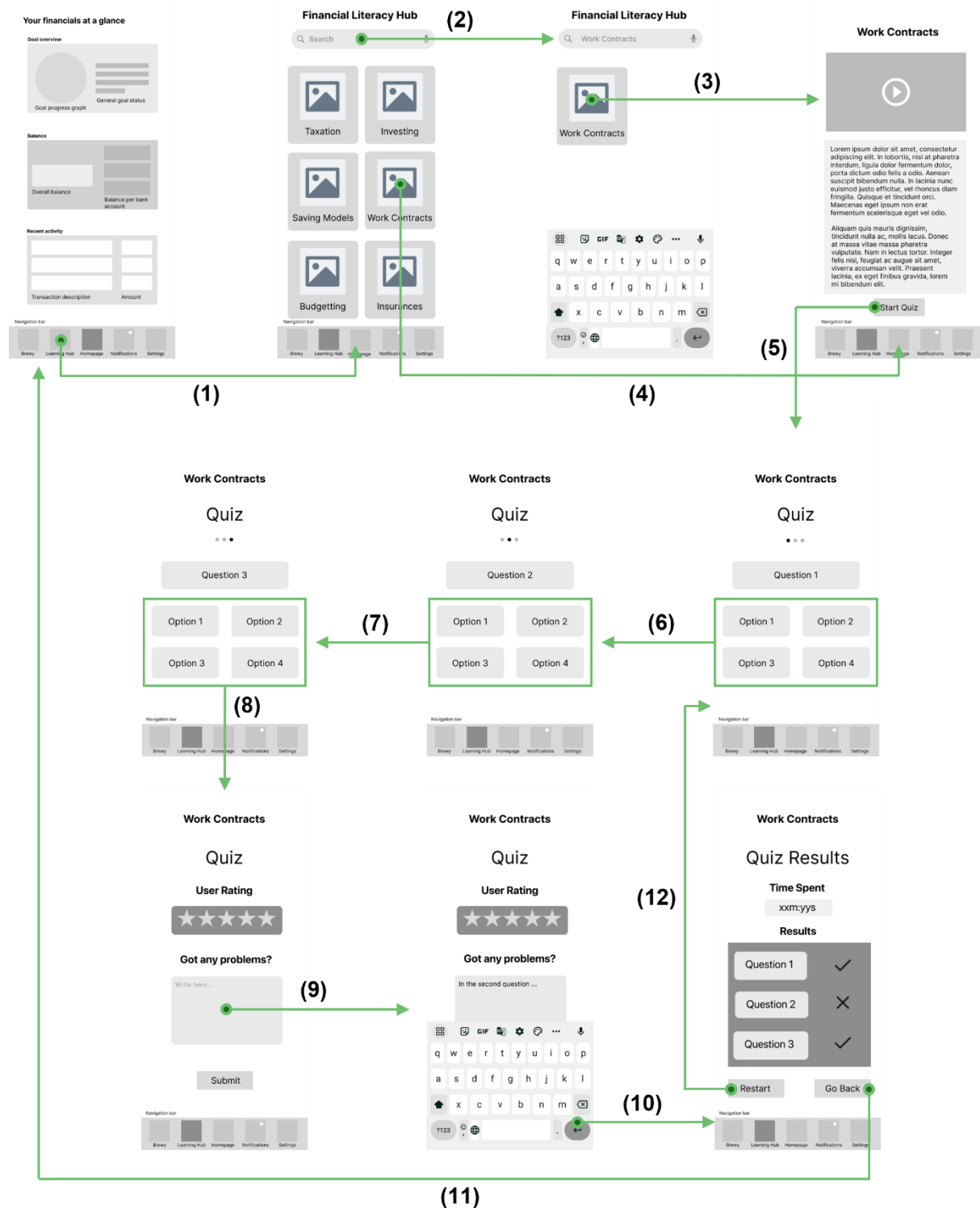
From the homepage, the user can press the Brewy icon on the navigation bar, moving onto the Brewy initial screen (1). Clicking on the text bar, a keyboard pops up (2), allowing them to write their prompt, in this case requesting an explanation of the 50-30-20 financial model. Hitting send shows an indicator suggesting the chatbot is “thinking” about its response (3), and after a few seconds, the expected explanation appears, clarifying the definition of the requested budgeting model (4).

Task 2: Remove an inactive Spotify subscription after receiving an automatic notification informing the user that they haven't listened to music for over 10 days



Starting off at the homepage, a pop-up shows up, alerting the user of a suggestion the app has determined, particularly that the app recommends the user cancels their Spotify subscription, as it deemed that the user has not been using the platform the last 10 days (1). The user then clicks on the "See details" button, taking them to a page detailing the potential saved money (2). On one hand, if the user clicks on "Cancel subscription" (3), a pop-up appears, requesting confirmation of the task, which can lead to a confirmation message if the user clicks "Yes" (5) that then automatically redirects to the homepage (7), or to the homepage itself, if the user clicks "Go back" (6). On the other hand, if the user chooses to "Keep active", clicking the associated button, they are redirected back to the homepage (4).

Task 3: Access a specific learning page (with video and article) about Work Contracts, and take a small quiz about the learned content



At the homepage, the user can access the financial literacy hub, by clicking the second icon present on the navigation bar **(1)**. This opens the hub's central page, showing some relevant topics, as well as a search bar. The user can either find the "Work Contracts" option on the available topics, or search for it **(2)**, selecting exclusively the desired topic. Clicking on either of them **[(3) and (4)]** takes the user to the Work Contracts content page, which includes a video explaining the topic, as well as a detailed article. At the lower part of this screen, a "Start quiz" option is present that, when tapped, starts the Work Contracts quiz **(5)**. As the user answers the various questions, selecting one of the available options, MoneyBrew moves to the subsequent questions **[(6), (7) and (8)]**, until the quiz review options appear. The user can attribute a rating to the quiz, and provide feedback by clicking on the text form **(9)**, after which they, finally, receive their results **(10)**. They can, at the end, choose to exit back to the homepage **(11)**, or repeat the quiz **(12)**.

Digested Heuristics Evaluation Results

During one of the phase 2 classes, the prototype for each of the three defined classes was analyzed and evaluated by fellow colleagues of two other groups. Each individual feedback was associated with one or more of the ten Nielsen's usability heuristics that guide the identification of usability problems, and its severity classified as cosmetic (level 1), minor (level 2), major (level 3) or catastrophic (4).

In total, our prototypes received twelve separate feedback items related to different heuristics, namely visibility of system status (type 1), user control and freedom (type 3), consistency and standards (type 4), recognition rather than recall (type 6), flexibility and efficiency of use (type 7), and help and documentation (type 10). In terms of severity, four of them fitted the major issues category, six of them were considered to be minor issues, and two cosmetic issues were also presented, with no catastrophic issues identified in our prototypes.

Major usability issues

We received four individual usability issues whose severity was classified as major. Two of these related to user control and freedom: the mandatory and forced requirement of evaluating the quiz after its completion on task 3, and the highlighted "Cancel subscription" button on the main task 2 action. Regarding this last problem, the feedback particularly explained that the user may feel inadvertently biased and compelled to cancel a subscription because this button is highlighted, in comparison with the opposite "Keep active" button.

The remaining two major issues were of the recognition rather than recall type, with one of them relating to the task 3 quiz results, in which no option to review specific question answers (correct or incorrect) at the end of the quiz was available, and the other concerning the AI chatbot's introduction/initial message (task 1), which was reported to be too generic, with no indication of the finance-specific theme of the conversations to be had.

Minor usability issues

Most issues identified in the heuristic evaluations conducted by our colleagues had minor severity. One visibility of system status issue was reported in relation to the task 3 quiz, and referred to the lack of clear navigation through the various questions and, indirectly, of the indication of how many questions are left to be answered at a given moment.

Three additional user control and freedom problems were identified in our prototypes: no option to pause or exit the quiz during its completion in task 3, no option to reverse the cancelation of the subscription in task 2, and lack of message timestamps and editing functionality, and absence of voice messaging and image support in the Brewy chatbot (task 1). This last issue was also reported as simultaneously fitting into the flexibility and efficiency of use type.

The remaining issue was a general issue of inconsistency when labeling buttons (“Yes” *versus* “Go back”, detected on one of the screens of the second task), which fits into the consistency and standards heuristics type.

Cosmetic usability issues

Tasks 1 and 2 were also reported to have one cosmetic issue each. The first task included a general help and documentation problem, exposing missing help icon and documentation to assist on the use of the artificial intelligence tool. Finally, the second task had a visibility of system status issue, in which it was unclear for how long the user still had access to their current subscription. Indeed, cancelation of these types of services is commonly not immediate; thus, the user should be informed of how long they can still use the service, if they so wish.

Non-reported feedback

Beyond the specific issues that were formally reported to us by the evaluating groups, we also identified some details in their navigation throughout the task prototypes that we hereby document.

Firstly, there was a generalized confusion when trying to access elements of our screens that were not related to the tasks at hand. For example, if one of the testing users tried to or accidentally accessed the settings screen via the navigation bar, they would enter a point-of-no-return, as that screen was not implemented in the context of any of the tasks. Moreover, most vocalized feedback related to visual elements that were considered to be inadequate or non-responsive, which, again, related to how the prototype focused on a low-fidelity visualization of screens that were exclusively relevant for the three selected functionalities.

However, an issue that we visually identified but was not reported to us in the written heuristic evaluations, and that we consider to be a relevant change for the final phase, is the overall lack of support for the “Back” button at each possible screen. In fact, the user should at any given moment be able to go back to the previous screen (whenever possible given the nature of the task), without compromising the state of the system nor the ongoing task without further confirmation by the user.

Corrections to Perform on Phase 3

Based on the two heuristic evaluation reports received from each of the two groups, several usability issues were identified, and those problems must be addressed on the third phase of this project. The proposed corrections detailed below are organized by heuristic type, with the severity of the respective problem clearly identified to facilitate prioritization of interventions.

User control and freedom (type 3)

Induced subscription cancelation (major issue)

As previously mentioned, the subscription cancelation screen presents a significant problem by highlighting the cancelation button, which may induce the user into canceling the subscription. Moreover, this issue may also inadvertently lead users into canceling that subscription unintentionally. Although our thought process when highlighting the “Cancel subscription” button was, indeed, to motivate the user into accepting the app’s suggestion, as it would most likely lead to a benefit in them saving money and achieving their financial goals, ultimately, we agree the user should be able to make a pondered and unbiased decision on taking or not upon that suggestion. As such, and to address this issue, a more balanced design will be implemented where both buttons (“Cancel subscription” and “Keep active”) have the same visual hierarchy. Additionally, an extra confirmation dialogue will be introduced, which will unequivocally detail the implications of the cancelation action, requiring an explicit validation to proceed and thereby ensuring the decision is informed.

Mandatory quiz rating system (major issue)

The financial literacy hub quiz rating system does not allow users to skip it, which violates the principle of user control. In the next phase, the option to skip the rating will be implemented, so that users are not forced to do so in order to end and exit the quiz.

Lack of options to pause or exit an ongoing quiz (minor issue)

The absence of options to pause or exit an ongoing quiz limits user control. This issue will be resolved by implementing a clearly visible pause button at the top of the screen, allowing users to save progress and resume later, as well as an exit button that enables abandoning the quiz at any time.

Missing timestamps and message editing in the chatbot (minor issue)

The inability to view the date and time of prompts sent to the chatbot, as well as of its responses, will be corrected by adding visible timestamps next to each message. However, we consider the lack of editing functionality to be a non-issue. Indeed, although these “messages” can be viewed as such, they are, in reality, prompts to an AI bot. Therefore, editing a prompt after it was sent does not change the subsequent response sent by Brewy, nor the remaining of the conversation that already occurred. As such, we will not make any corrections regarding message editing. However, since some users have seemingly mistaken this feature as a person-to-person conversation, rather than an AI chatbot, we will highlight in the help and documentation for this functionality that further clarification of a response through new prompts, rather than prior message editing, is intended.

Unclear reversal of subscription cancelation (minor issue)

After confirming a subscription cancelation, there is no clear way to reverse this action. To solve this problem, a 24-hour grace period will be introduced, during which users can undo the cancelation through a dedicated section in the application settings.

Recognition rather than recall (type 6)

Inability to review detailed quiz results (major issue)

As previously stated, the quiz results page only indicates whether answers were correct or incorrect, without allowing review of the questions and answer options. This limitation will be corrected by implementing a detailed review screen that presents each question, all answer options, the user's response, and the correct answer, accompanied by pedagogical explanations where applicable.

Overly generic chatbot introductory message (major issue)

The introductory message of the AI chatbot was considered too generic, failing to clearly indicate that Brewy is, indeed, a financial assistant. This issue will be resolved by reformulating the initial message to include a specific mention of its finance-related range, as well as further clarification in the feature's help button (further detailed in another issue below).

Visibility of the system status (type 1)

Missing quiz progress indicator (minor issue)

Currently, the quiz page does not indicate how many questions remain to be answered. This gap will be filled by adding a visible progress indicator (e.g., "Question 3 of 10") at the top of the screens presenting each quiz question.

Ambiguity regarding access after subscription cancelation (cosmetic issue)

After confirming a subscription cancelation, it is unclear whether the user retains access to the service until the end of the billing cycle or if access is immediately revoked. This ambiguity will be eliminated by displaying an explicit confirmation message that clearly indicates when access to the service will end (e.g., "Your subscription will be terminated on December 15, 2025. You have access to the service until that date.").

Consistency and standards (type 4)

Inconsistency in confirmation options (minor issue)

The subscription cancelation confirmation page presents two inconsistent options, "Yes" and "Go back". This issue will be resolved by standardizing the options to "Yes, cancel the subscription" and "No, keep the subscription", maintaining consistency with application standards and avoiding confusion between choices and system actions.

Flexibility and efficiency of use (type 7)

Limited message types in chatbot interaction (minor issue)

The inability to send voice messages or images to the Brewy chatbot limits user interaction options. As such, in the next phase, functionality for sending voice messages and images (for example, photographs of invoices or receipts) will be implemented in the prototype, thereby significantly expanding the assistant's capabilities and making interaction more natural and efficient.

Help and documentation (type 10)

Absence of help button in chatbot (cosmetic issue)

The Brewy chatbot does not have a help button explaining its functionalities and capabilities. To address this issue, we will add an information button to the chatbot interface that, when activated, presents a brief description of the assistant's capabilities, examples of frequently asked questions, and usage suggestions, facilitating user understanding of how to make the most of this feature.

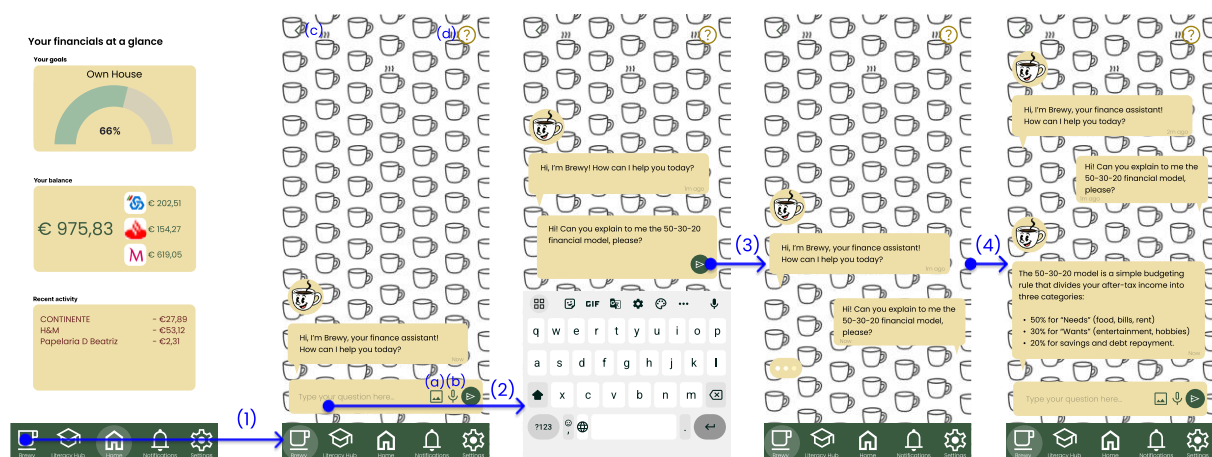
Part III: Hi-fi Prototype and User Evaluation

Prototype's Wireflow

After defining the necessary corrections in Phase II and re-analyzing the low-fidelity prototype behaviors and functionalities, we designed a functional and high-fidelity prototype. This prototype elucidates the expected pathway users are expected to follow in order to accomplish the aforementioned tasks and is designed to be more user-friendly.

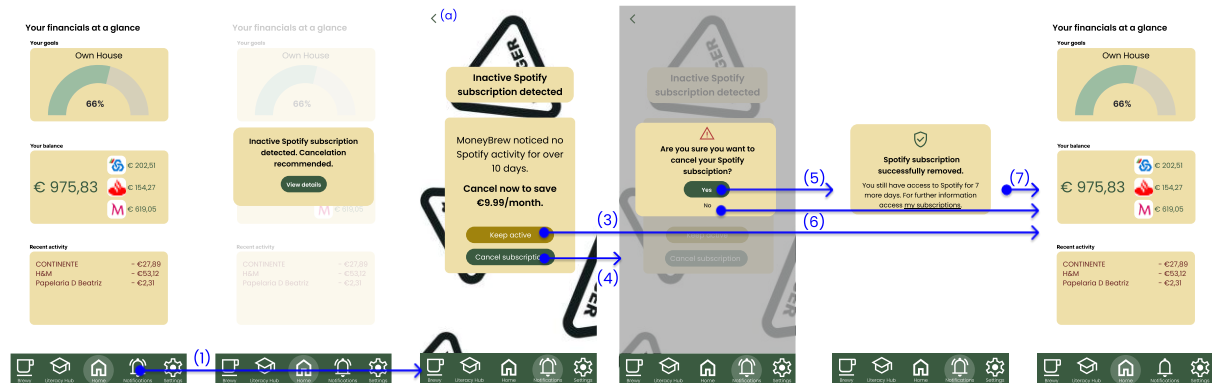
Accordingly, the present chapter details the updated wireflow of each individual task. As in Phase II, it was decided to start each wireflow on the homepage screen, as a central point of the app's navigation. However, the execution could begin from anywhere on the app for tasks 1 and 3, as the relevant screens are easily accessible via a consistently-present navigation bar, and from clicking the notification for task 2, independently of where the user was within the app at the time they received that alert. The live version can be accessed through [here](#).

Task 1: Ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model is



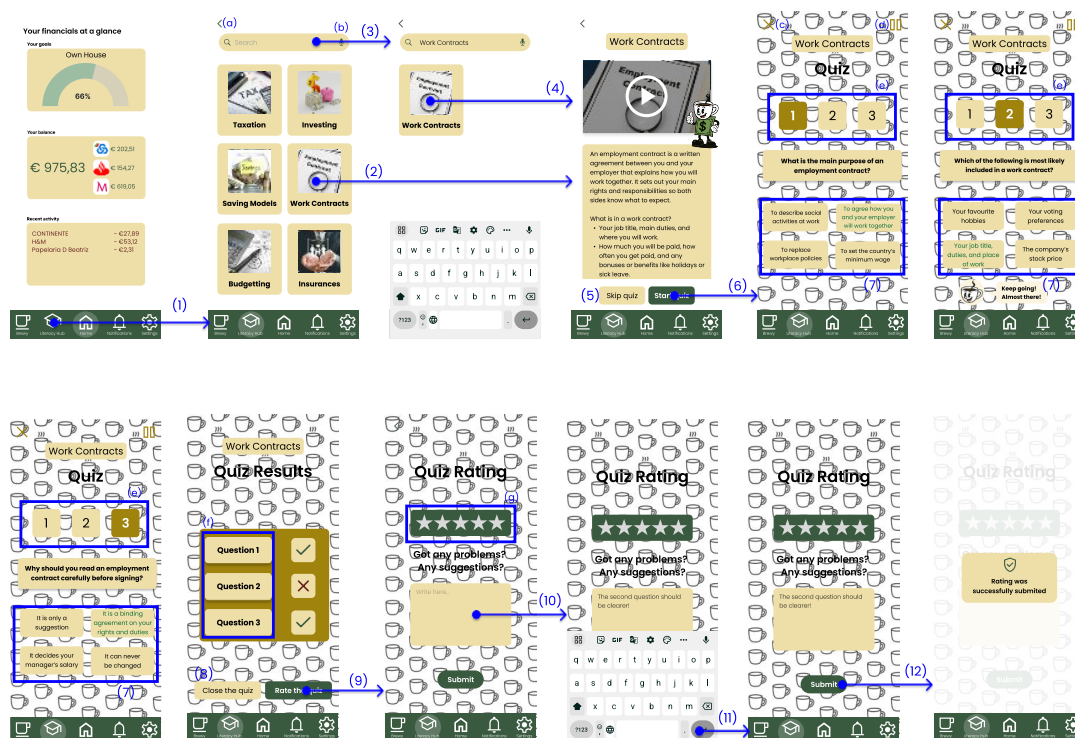
From the homepage, the user must tap on the Brewy icon on the navigation bar, after which they are redirected to the AI chat screen (1). There, the user sees several new icons. The image icon (a) gives the user the opportunity to upload images to our agent, and the microphone icon (b) enables the user to send voice messages to Brewy. The “Go back” icon (c) gives the user a way back to the screen they were before, and the help icon (d) redirects them to a screen explaining how Brewy can be helpful. On this screen, the user must click on the text bar to write the wanted message (2), and then hit “Send” (3). After some time, during which our bot is “thinking”, it gives an answer (4) explaining better the financial model.

Task 2: Remove an inactive Spotify subscription after receiving an automatic notification informing the user that they have not listened to music for over 10 days



On the homepage, and after receiving the notification pop-up, the user can either interact with it (2) or use the navigation bar (1) to see notifications. On the screen with the notification details, we also added a “Go back” button (a), so that the user has more freedom and can better control their flow in our application. Once the user accesses the notification, they can choose between the “Keep active” (3) and “Cancel subscription” options (4). By choosing the first, the user goes directly back to the homepage, but by choosing the other, the user sees a confirmation page to avoid any unintended action. On the confirmation page, the either confirms the action (5) or cancels it (6). The first option redirects to a success message and then to the homepage (7), and the second redirects directly back to the homepage.

Task 3: Access a specific learning page (with video and article) about Work Contracts, and take a small quiz about the learned content



Starting at the homepage, the user must pick the “Literacy Hub” icon **(1)**. On the Literacy Hub page, the user sees icons **(a)** and **(b)**. Icon **(a)** is a “Go back” button to the previous screen the user was on, and the icon **(b)** is a microphone button so the user can search by voice. Still on this page, the user can either directly scroll and pick the wanted content **(2)** or can use the search tool **(3)** and then select the content **(4)**. Once the user is done watching the video and reading the article, they can choose between “Skip quiz” **(5)** that redirects back to the initial literacy hub page and “Start quiz” **(6)**.

On the quiz page new icons are introduced, icons **(c)** and **(d)**. Icon **(c)** gives the power to the user to close the quiz any time, and icon **(d)** gives the chance to pause and then resume the quiz. The user can also navigate freely between questions using the navigation **(e)** and, in order to finish the quiz, the user must pick an option out of the available **(7)**. On the results page, the user can review the questions **(f)** and check which was the correct answer. Also on this page, the user can close the quiz without giving a review **(8)** or rate the quiz as intended **(9)**. On the quiz rating page, the user can rate **(g)** and leave suggestions by clicking on the text bar **(10)**. Then, after submitting the quiz review **(12)**, there is a confirmation page and the user goes automatically back to the literacy hub initial page.

User Evaluation Protocol

Objectives

In order to evaluate the quality of the user interface created for the MoneyBrew mobile application, an evaluation protocol was developed and carried out in testing sessions with a selected group of users. These individuals were identified as being representative of the main target audience of the app, being adults in employment (and thereby, earning) age, while simultaneously exhibiting variety in terms of their technical expertise. With these evaluation sessions, we aimed to observe and assess their first contact and interactions with MoneyBrew’s interface, in the context of three previously identified, concrete tasks.

The sessions consisted of assigning the user to ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model consists of, removing an inactive Spotify subscription after receiving an automatic notification informing the user that they haven’t listened to music for over 10 days, and accessing a specific learning page (with video and article) about “Work Contracts” and taking a small quiz about the learned content. The users were asked to attempt to execute and fully complete these tasks while using a high-fidelity mockup of the application that supported the interfaces required for each of them. Measurements, such as “misclicks” (accidental or voluntary) made by the users, the time it took to complete each task (and individual actions within each of them) and the user’s overall opinion and considerations about the tasks and the overall interface were registered, in order to posteriorly perform an analysis of the results and answer the following questions:

1. Can users complete the three defined tasks successfully?
2. Was the completion achieved intuitively, simply and easily?
3. What were the challenges or the mistakes the users experienced or did when attempting to complete the tasks?
4. How many attempts were required for the users to successfully complete the tasks?
5. Was the overall time of completion of the tasks as expected?

6. How do users classify their overall experience with doing each task?
7. How do users classify their overall experience while interacting with the app's interface?
8. Do the users see a benefit in using the app?
9. How likely is it that the users would use the app in the future?
10. What are the details that must be clarified or changed (e.g., provide more help or context guides, etc.)?

With the answers to these questions, we were able to objectively evaluate if the user interface for MoneyBrew is of good quality, providing a pleasant and satisfactory user experience, and identify potential areas for improvement.

Users

MoneyBrew is targeted for financially independent individuals who desire to manage their personal finances in a simple and intuitive way. For this reason, it is expected that the main users of the application are earning people, who can be from various ages groups, backgrounds, and lifestyles. The MoneyBrew concept was developed to serve both the young adult and the senior, the working student and the retired individual, the early school leaver and the postgraduate, the expert in technology and the traditionalist.

Due to this diverse scope, we had to make sure that individuals who were representative for each of these groups took part of the evaluation. As a result, it was defined that we were to include as the sample for our experiment a total of 10 individuals: 3 of them being younger individuals, including a student, a full-time worker and a working student, four adults ranging from 25 to 45 years old with different educational backgrounds and lifestyles, and 3 older individuals with averagely the same overall characteristics (employed, with lower tech-savviness). This distribution was established because it was predicted to be proportionally equivalent to the potential distribution of the end user population for the application.

Taking into consideration the context in which the project was being developed, the participants for the evaluation were selected among our relatives, colleagues and friends, also allowing for an easier achievement of the desired distribution and variance of our recruits, but, nevertheless, implying a possible positive bias towards our project.

Method

In order to obtain reliable and unbiased results in terms of how the evaluation session was carried out, a comprehensive and strict methodology was developed and meticulously followed by the hosts.

Each session consisted of three primary moments. The first one was intended to obtain some relevant general information about the user, such as age, educational background and previous experience with technologies, in addition to introducing them to the MoneyBrew application and explaining extensively the procedures of the session. With these guidelines in mind, each session started with the reading of the following text, which explained in detail the session in its entirety, presented the Maze environment in which the application's user interface would be accessible, and introduced the concept of MoneyBrew, followed by questioning the participant if any additional information was needed or something was not

clear. The English-translated version of this introductory text, as well as its original version (in Portuguese) were as follows:

English version

“Please listen carefully to the following information.

This test aims to evaluate the quality of the interface developed for the mobile application MoneyBrew by observing, in real-time, the first contact potential users have with it. The test will be divided into three distinct stages. The first will consist of an initial questionnaire to record some of your general characteristics and your experience with technologies similar to the one you will be interacting with. The second stage will be based on three tasks that you should attempt to complete using a simulation of the application. The objective of each task will be presented beforehand, and it will be up to you to figure out how to complete it successfully and autonomously. The only questions you may ask during the task execution period concern the translation or meaning of terms that are in English. After each task, you will answer brief questions regarding your experience with it. Once all three tasks are finished, a final questionnaire will be conducted, this time regarding your overall experience and considerations about the application's interface as a whole.

To perform the tasks, the application simulation will be presented on the computer via the Maze platform. There, you will see a screen where the objective of the first task is described on the left, with a blue button at the bottom named "Let's start", and the application simulation on the right. By clicking the button, the simulation will become accessible. The way you interact with the simulation is entirely similar to a conventional mobile app, based on screen taps and scrolling, this time simulated with the mouse. When you successfully complete a task, Maze will automatically notify you, directing you to a screen with the inscription "Great!" and a "Continue" button. Clicking it will take you to the next task, and so on. After the last task, you will be directed to a thank-you screen.

It is important to emphasize that not all of the application's interface is implemented. Therefore, if during the test you attempt to access a screen or execute a task that is not implemented, you should click the back button or choose another alternative that you are able to select on the screen. Additionally, whenever the tasks involve searching for or submitting text, you do not need to type anything; by clicking on the search or text box, the text you would need to write will appear pre-filled, and you will only have to submit it as if you had just finished writing.

To provide context for the application itself: MoneyBrew is an intelligent personal finance management platform. It allows users to easily, quickly, and intuitively manage their bank accounts, track their expenses and earnings, make and monitor investments, and receive useful tips and information, among many other features—helping busy people who are concerned about their financial well-being.

If you have any questions regarding this information, you may ask them now. (...)

We will now begin the test, and we ask for your thoughtfulness and honesty over the next few minutes. We emphasize that it is not you, the user, who is being evaluated, but rather the interface. Finally, should you wish to do so, you may terminate the test at any time.”

Portuguese version

“Ouça com atenção a seguinte informação.

Este teste tem como objetivo avaliar a qualidade da interface desenvolvida para a aplicação móvel MoneyBrew, através da observação em tempo real do primeiro contacto de potenciais utilizadores com a mesma.

O teste será dividido em três momentos distintos. O primeiro será a realização de um questionário inicial para registar algumas das suas características gerais, bem como da sua experiência com tecnologias semelhantes àquela com a qual irá interagir. O segundo momento basear-se-á em três tarefas que deverá tentar completar usando uma simulação da aplicação. O objetivo de cada uma das tarefas será previamente apresentado, cabendo-lhe a si descobrir como completá-las com sucesso de forma autónoma. As únicas questões que poderá fazer durante o período de realização da tarefa serão relativas à tradução e ao significado de termos que se encontrem em inglês. Após cada tarefa, responderá a breves perguntas relativas à sua experiência com a mesma. Depois de concluídas as três tarefas, será realizado um último questionário, desta vez relativo à sua experiência no geral, e a considerações acerca da interface da aplicação como um todo.

Para executar as tarefas, será apresentada a simulação da aplicação no computador, na plataforma Maze. Aí, observará um ecrã onde, à esquerda, está descrito o objetivo da primeira tarefa e um botão azul em baixo com a expressão “Vamos começar”, e à direita encontra-se a simulação da aplicação. Clicando no botão, a simulação tornar-se-á acessível. O modo de interagir com a simulação é em tudo semelhante a uma aplicação convencional para telemóvel, baseando-se em toques no ecrã e em scrolling, desta vez simulados com o mouse. Quando terminar com sucesso uma tarefa, o Maze avisar-lhe-á automaticamente, encaminhando-lhe para um ecrã com a inscrição “Muito bom!” e com um botão “Continuar”. Clicando nele, será encaminhado para a tarefa seguinte, e assim sucessivamente. Após a última tarefa, será encaminhado para um ecrã de agradecimento.

É importante salientar que nem toda a interface da aplicação se encontra implementada. Logo, caso durante o teste tente aceder a um ecrã ou executar uma tarefa não implementada, deverá clicar no botão de voltar atrás, ou escolher outra alternativa que consiga selecionar no ecrã. Além disto, sempre que as tarefas implicarem a pesquisa ou submissão de texto, não necessita de escrever qualquer texto; ao clicar na caixa de pesquisa ou escrita, o eventual texto que necessitaria de escrever já irá aparecer pré-preenchido, e apenas terá de submeter, como se tivesse acabado de escrever.

Para contextualizar a aplicação em si, MoneyBrew é uma plataforma de gestão inteligente de finanças pessoais, permitindo aos seus utilizadores, de forma fácil, rápida e intuitiva, gerirem as suas contas bancárias, controlarem as suas despesas e ganhos monetários, fazerem e controlarem investimentos, receberem dicas e informações úteis, entre muitas outras funcionalidades, ajudando pessoas ocupadas, mas preocupadas com o seu bem estar financeiro.

Caso tenha alguma dúvida sobre estas informações, pode agora colocar. (...)

Iremos então iniciar o teste, e pedimos ponderação e honestidade durante os próximos minutos. Ressalvamos que não é o utilizador que está a ser avaliado, mas sim a interface. Por último, se assim o desejar, poderá dar o teste como terminado a qualquer momento.”

At the end of this first part, a quick questionnaire was made to the participant, in order to gather some relevant information about their general characteristics. It consisted of the following questions (translated into English):

1. What is your age?
2. What is your highest level of completed education?
3. What is your current professional status (student, worker, etc.)?
4. On a scale of 1 to 5, what is your level of experience with mobile technologies?
5. Have you ever used a financial management app? If so, how often do you use it?

The second part of the sessions consisted on the execution by the participant of the three defined tasks. Before each task, a text was read presenting the functionality of the application related to the task and explaining its objective, where the user would start within the application interface (e.g. the homepage), and when the task would be considered to be completed with success. The introductory texts, both their versions in Portuguese (used in the sessions) and the respective English translations, for each of the tasks are presented below.

Task 1 Introductory Text

English version

“One of the most notable features of the MoneyBrew application is the presence of an AI chatbot called Brewy, which can answer any question asked by the user or provide tips and explanations regarding personal finance or economic and fiscal matters. In this first task, you will start on the application's homepage and must access Brewy; then, you should request an explanation from it about the “50-30-20” personal finance management model. The task will be considered successfully completed when you receive the respective explanation from Brewy.”

Portuguese version

“Uma das funcionalidades mais notáveis da aplicação MoneyBrew é a presença de um chatbot de inteligência artificial chamado Brewy, que poderá responder a qualquer pergunta colocada pelo utilizador ou fornecer dicas e explicações relativas a finanças pessoais ou assuntos económicos e fiscais. Nesta primeira tarefa, irá começar na página inicial da aplicação e deverá aceder ao Brewy; posteriormente, deverá requerer-lhe uma explicação acerca do modelo “50-30-20” de gestão de finanças pessoais. A tarefa será considerada concluída com sucesso quando receber a respetiva explicação por parte do Brewy.”

Task 2 Introductory Text

English version

“In addition to allowing you to manage your expenses, such as monthly subscriptions, the MoneyBrew application also monitors how frequently you use the services associated with them. In this second task, you will begin on the application's homepage and receive a notification alerting you to the fact that you have not accessed the Spotify platform during the last ten days, despite subscribing to the service. From there, you must follow the recommendation mentioned in that notification. The task will be considered successfully completed once that action is finished.”

Portuguese version

“A aplicação MoneyBrew, além de permitir a gestão das suas despesas, como por exemplo assinaturas mensais, também controla a frequência de utilização dos serviços associados às mesmas por parte do utilizador. Nesta segunda tarefa, irá começar a mesma na página inicial da aplicação, e receberá uma notificação alertando para o facto de não ter acedido à plataforma Spotify durante os últimos dez dias, apesar de subscrever esse serviço. A partir daí, deve seguir a recomendação mencionada nessa notificação, sendo a tarefa considerada completa com sucesso quando tal for concluído.”

Task 3 Introductory Text

English version

“A good level of financial literacy, combined with the adoption of good personal finance management practices, is essential for a user's economic well-being; however, the former is often scarce. In this sense, MoneyBrew provides a platform where the user can learn about various financial matters in a simple and intuitive way. In this third and final task, starting from the application's homepage, you must access the Financial Literacy Hub and navigate to the section on "Work Contracts." When the specified content appears, you must take a quiz on the subject and then provide feedback using the star rating system. After completing these two actions, the task is considered successfully finished. It is worth noting that the answers given during the quiz, as well as the number of stars awarded at the end, do not affect the success or failure of the task's completion.”

Portuguese version

“Um bom nível de literacia financeira, aliado à adoção de boas práticas de gestão pessoal de finanças, são essenciais para o bem-estar económico do utilizador; porém muitas vezes esta primeira é escassa. Nesse sentido, MoneyBrew fornece uma plataforma onde o utilizador poderá aprender de forma simples e intuitiva acerca de diversos assuntos financeiros. Deste modo, nesta terceira e última tarefa deverá, partindo da página inicial da aplicação, aceder ao “Hub” de literacia financeira, e aceder à secção sobre “Contratos de Trabalho”, ou “Work Contracts”. Quando aparecer o conteúdo especificado, deverá responder a um quiz acerca do assunto, devendo, de seguida, dar o respetivo feedback através do número de estrelas. Após a conclusão destas duas ações, a tarefa é considerada terminada com sucesso. Vale ressaltar que as respostas dadas durante o quiz, bem como o número de estrelas atribuídas no final, não interferem no sucesso ou insucesso da conclusão da tarefa.”

After the explanation of the purpose of the task, the user is left with the simulation of the application to interact with and try to successfully complete the respective task. During this time, registrations were performed by the host, such as the number of unexpected clicks (representing taps) on the screen, particularly mistakes (such as going through the wrong path or misclicks), and the whole process was timed and recorded for future analysis (more on this on the “Measures” section), using the Maze platform. There was a previously defined time limit per task, which was not disclosed to the participant, in order to not potentially bias their behavior when performing the task. It consisted of 1 minute for Task 1, 1 minute for Task 2, and 2 and a half minutes for Task 3. If the time limit was exceeded, that attempt was deemed as unsuccessful.

After each task, the participants were asked a series of questions, whereby they were asked to give their ratings and opinions about their experience with the task, the interface elements associated with it, the overall visual appearance, what were their thoughts about the functionality, their performance while doing the task and the motives of verifiable mistakes. The presented questions are listed below.

1. What was the reason you clicked here when you were doing this? (Asked for each misclick.)
 - a. In the case of the second task, if you chose to keep the subscription, why did you do so?
2. On a scale of 1 to 5, how efficient do you consider your execution of the task, where 1 implies it required many actions on your part, and 5 that it was extremely efficient, having been completed in very few clicks or taps?
3. On a scale of 1 to 5, how fast or slow was your execution of the task for you and according to your expectations, where 1 implies it was excessively slow, and 5 that it was extremely fast?
4. On a scale of 1 to 5, how easy or difficult did you find this task, where 1 implies it was very complex, and 5 to be quite simple?
5. On a scale of 1 to 5, how would you rate your experience with the task, where 1 is very negative, and 5 is very positive?
6. On a scale of 1 to 5, how do you visually evaluate the app's appearance, specifically regarding the elements you interacted with for this task, where 1 is not at all appealing, and 5 is very appealing?
7. Assuming you were to install this application, would you use this particular feature?
8. Do you have any suggestions or comments related to the task?

The third and final moment of the session consisted on the application of a questionnaire in which the participant is asked to give their considerations about the MoneyBrew system as a whole, more specifically about its visual appearance, the user's experience while interacting with it and its appeal and utility, by responding to the questions below.

1. On a scale of 1 to 5, how do you visually evaluate the general appearance of the application, where 1 is not at all appealing and 5 is very appealing?
2. On a scale of 1 to 5, how would you rate your overall experience with the application, where 1 corresponds to very negative and 5 to very positive?
3. Based on the general objective of the application presented to you, the features you had the opportunity to test, and your general interest in using technology to help reach your financial goals, would you use this application? If not, why?
4. Do you have any suggestions or comments related to the application in general?

Tasks

The three tasks that were defined during phase II of the project were the ones submitted to testing during the user evaluation sessions.

Regarding the first one, its objective was to ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model is. To achieve this, the participant, who started on the

MoneyBrew homepage, had to find the way to access Brewy, and then, enter the correct prompt in order to be provided with the desired answer. The task was considered completed with success after Brewy responded with the requested information about the 50-30-20 financial model.

The second task had the user remove an inactive Spotify subscription after receiving an automatic notification informing the user that they hadn't listened to music for over 10 days. In summary, the user, who, again, started on the homepage, received a pop-up notification alerting to the fact that they hadn't used Spotify for over 10 days despite subscribing to its service, so it recommended the cancelation of the subscription. From that, the user had to proceed with the cancelation process, with the task being completed when that was fulfilled.

Regarding the third and final task, it was proposed to the user to access a specific learning page about Work Contracts, and to take a small quiz about that content. Similarly to the previous tasks, the participant started on the homepage, where they had to find how to access the Financial Literacy Hub. From there, they were required to find and enter the section about "Work Contracts" and, after browsing through the content, start the quiz. After all the three multiple choice questions of the quiz had been answered, the user had to give feedback about the content and the quiz. The task was considered to be fulfilled if both tasks, completing and evaluating the quiz, were finalized.

Measures

For each one of the three defined tasks, several data points related to the user interactions with the app interface were collected from the evaluation sessions, in order to measure the efficacy, efficiency and satisfaction, as defined in the Phase II of the project.

In relation to the first task, we measured the time it took to complete the full task, and the number of accidental and voluntary misclicks (clarified after the end of the task in the post-task questionnaire), along with the respective justifications, as well as the rating scores requested on the post-task questionnaire. The task is considered to be completed if its goal was achieved within the time limit previously stipulated. As defined in the previous phase of the project (but reviewed when considering the small size of our sample), the associated success thresholds consisted of 80% of users successfully completing the task, an average number of user taps of no more than 5, an average completion time of up to 2 minutes, and the average rating (considering all scores) on the post-task questionnaire to be about 4 or 5 out of 5.

Regarding the second task, we informed each user that they had to follow the suggestion that the app offered in the notification, without clarifying if that meant to cancel or keep the Spotify subscription. This allowed us to understand whether the suggestion wording is sufficiently clear for the user to follow through with it. Similar to the first task, we recorded the full time of task execution and the misclicks each user did while carrying it out. Of note, we established beforehand that if the user ended up not cancelling the subscription, therefore choosing to keep it active, we were to consider this task to be completed without success, and inquired the user on the reason why they did so, to discriminate between task misunderstanding, lack of comprehension of the meaning of the suggestion, or accidental misclicking of the "Keep active" button. The corresponding success thresholds were defined as 80% of users successfully completing the task, the average number of user taps to be no more than 5, the average completion time to be under 2 minutes, and the average rating on the post-task questionnaire to be about 4 or 5 in 5.

As for the final task, it was considered to be fully completed when the quiz was finished and the rating submitted, partially completed if the quiz was completed, but the user did not rate it, and failed when the quiz was not finished (and, consequently, neither was its rating). We also took note of the time the user took to carry out the task, and the number of unexpected clicks. The associated success thresholds consist of 80% of users successfully completing the task, an average number of user taps to be no more than 12, an average completion time up to 5 to 10 minutes, and an average rating on the post-task questionnaire to be about 4 or 5 in 5.

Results

Sample characterization

Our user evaluation sample is comprised of 10 individuals aged 20 to 60 years (median 32 years), mostly secondary- or college-educated, employed, and mobile-tech-savvy, who primarily never used a financial management application. It, therefore, reflects MoneyBrew's target user base, consisting of busy, employed adults concerned with personal financial control, while simultaneously presenting a mix of experienced and inexperienced individuals, essential for performing a comprehensive evaluation of interface intuitiveness. A detailed characterization of the sample is presented on Table 1.

Table 1. Demographic characterization of the MoneyBrew evaluation user sample.

User evaluation sample (n = 10)	
Age, median (range), years	32 (20 – 60)
18 ≤ Age ≤ 25, n (%)	3 (30)
26 ≤ Age ≤ 40, n (%)	4 (40)
Age ≥ 41, n (%)	3 (30)
Mode, years	20
Quartile 1 (25 th percentile), years	23.00
Quartile 3 (75 th percentile), years	48.75
Highest completed education level, n (%)	
Primary education	2 (20)
Secondary education	5 (50)
Bachelor's degree	3 (30)
Employment status, n (%)	
Student	1 (10)
Working student	1 (10)
Employed	8 (80)
Experience with mobile technologies (Scale 1 – 5), n (%)	
Score 1	0 (0)
Score 2	1 (10)
Score 3	1 (10)
Score 4	3 (30)
Score 5	5 (50)
Mode	5
Quartile 1 (25 th percentile)	3.75
Quartile 3 (75 th percentile)	5.00
Prior usage of similar apps, n (%)	
Never used	6 (60)
Previously used	4 (40)

Statistical analysis

Statistical analyses of the data collected during user evaluations were performed using the SPSS Statistics software, version 30.0 (IBM), and graphical representations designed using the GraphPad Prism software, version 8.0 (GraphPad Software, Inc).

Descriptive statistics

Task 1: Ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model is

In our evaluation, nine users completed the first task (90%), with only one user failing to do so (10%). The mean completion time was 22.79 seconds, with a standard deviation of 10.748 seconds. Throughout task execution, there was only one intentional misclick, from a user who scrolled through the homepage in search of the Brewy page. Regarding the evaluated ratings, users provided positive feedback, with all median classifications being 5 out of 5, with the exception of the visual appearance classification (4/5 median classification). Of the individuals who finished this task, eight of them reported that they would use its functionality, assuming they had installed the app. The complete descriptive results are presented below.

Table 2. Task 1 descriptive statistics results.

Task 1 descriptive statistics (n = 10)	
Fully completed the task, n (%)	
Yes	9 (90)
No	1 (10)
Task completion time, mean (standard deviation), seconds	
Quartile 1 (25 th percentile)	13.50
Quartile 2 (50 th percentile, median)	19.00
Quartile 3 (75 th percentile)	33.10
Total misclicks, n (%)	
Irrelevant misclicks	0 (0)
Intentional misclicks	1 (100)
Accidental misclicks	0 (0)
User ratings (Scale 1 – 5)	
Efficiency, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	0 (0)
Score 5	9 (90)
N/A	1 (10)
Mode	5
Quartile 1 (25 th percentile)	5.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Speed, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	3 (30)
Score 5	5 (50)
N/A	1 (10)
Mode	5
Quartile 1 (25 th percentile)	4.00

Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Ease of execution, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	0 (0)
Score 5	9 (90)
N/A	1 (10)
Mode	5
Quartile 1 (25 th percentile)	5.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Visual appearance, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	6 (60)
Score 5	2 (20)
N/A	1 (10)
Mode	4
Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	4.00
Quartile 3 (75 th percentile)	4.50
Overall experience, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	1 (10)
Score 5	8 (80)
N/A	1 (10)
Mode	5
Quartile 1 (25 th percentile)	5.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Would use the task's functionality, <i>n</i> (%)	
Yes	8 (80)
No	1 (10)
N/A	1 (10)

Task 2: Remove an inactive Spotify subscription after receiving an automatic notification informing the user that they haven't listened to music for over 10 days

All 10 users of our sample completed this second task, with an average completion time of 15.50 seconds (standard deviation, 6.460 seconds). A total of two misclicks were recorded, with one being accidental, and the other corresponding to a tester who believed they had to click the Notifications icon on the navigation bar to access the notification. User ratings were all uniformly positive, with all median scores, except the visual appearance classification (4.50), being the maximum of 5.00. All users mentioned that, if they were to install the app, they would use this smart notification functionality. All task 2 results are presented in Table 2.

Table 3. Task 2 descriptive statistics results.

Task 2 descriptive statistics (n = 10)	
Fully completed the task, n (%)	
Yes	10 (100)
No	0 (0)
Task completion time, mean (standard deviation), seconds	
Quartile 1 (25 th percentile)	9.56
Quartile 2 (50 th percentile, median)	15.80
Quartile 3 (75 th percentile)	19.03
Total misclicks, n (%)	
Irrelevant misclicks	1 (50)
Intentional misclicks	1 (50)
Accidental misclicks	0 (0)
User ratings (Scale 1 – 5)	
Efficiency, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	1 (10)
Score 5	9 (90)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	5.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Speed, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	0 (0)
Score 5	10 (100)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	5.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Ease of execution, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	1 (10)
Score 5	8 (80)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	4.75
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Visual appearance, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	4 (40)
Score 5	5 (50)
N/A	0 (0)
Mode	5

Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	4.50
Quartile 3 (75 th percentile)	5.00
Overall experience, n (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	1 (10)
Score 5	9 (90)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	5.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Would use the task's functionality, n (%)	
Yes	10 (100)
No	0 (0)
N/A	0 (0)

Task 3: Access a specific learning page (with video and article) about Work Contracts, and take a small quiz about the learned content

The final task was completed, partially or fully, by all elements of the testing sample. Indeed, it is important to note that two individuals did not perform the quiz rating step; however, since this only corresponded to a small step in the context of the overall task, which also included accessing the “Work Contracts” section of the financial hub, as well as completing the associated quiz, we included the data collected from these two testers in this task’s analyses.

The ten individuals took, on average, 22.79 seconds to carry out the task, with a standard deviation of 35.579 seconds. In terms of misclicks, a total of five were observed: two that were considered to be irrelevant (misclicks due to task misunderstanding), two accidental (unintended clicks on the settings and on the homepage), and one intentional, in which the user attempted to access the financial hub by hitting the settings icon. Furthermore, classification scores were generally high, with the visual appearance rating, similarly to the prior two tasks, receiving the lowest median classification (4.50). Nevertheless, all testers mentioned they would use the financial hub, given they would install the MoneyBrew application.

Table 4. Task 3 descriptive statistics results.

Task 3 descriptive statistics (n = 10)	
Fully completed the task, n (%)	
Yes [†]	10 (100)
No	0 (0)
Task completion time, mean (standard deviation), seconds	
Quartile 1 (25 th percentile)	29.40
Quartile 2 (50 th percentile, median)	58.50
Quartile 3 (75 th percentile)	88.12
Total misclicks, n (%)	
Irrelevant misclicks	2 (40)
Intentional misclicks	1 (20)
Accidental misclicks	2 (40)

User ratings (Scale 1 – 5)	
Efficiency, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	4 (40)
Score 5	6 (60)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Speed, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	2 (20)
Score 5	7 (70)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Ease of execution, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	0 (0)
Score 4	2 (20)
Score 5	8 (80)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	4.75
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00
Visual appearance, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	3 (30)
Score 4	2 (20)
Score 5	5 (50)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	3.00
Quartile 2 (50 th percentile, median)	4.50
Quartile 3 (75 th percentile)	5.00
Overall experience, <i>n</i> (%)	
Score 1	0 (0)
Score 2	1 (10)
Score 3	0 (0)
Score 4	3 (30)
Score 5	6 (60)
N/A	0 (0)
Mode	5
Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	5.00
Quartile 3 (75 th percentile)	5.00

Would use the task's functionality, <i>n</i> (%)	
Yes	10 (100)
No	0 (0)
N/A	0 (0)

¹ Two users (20%) completed the task in a partial manner, by not conducting the quiz rating, due to a misunderstanding of part of the tasks' instructions; because this step was only a small part of the full task, the data collected from the evaluation of these two cases was included in the statistical analyses.

Overall application evaluation

As aforementioned, we also collected user opinions regarding the app as a whole, particularly its visual appearance, the overall experience of interacting with its prototype, and whether the users would use it, if it was available. The results were very positive, with median scores of the two ratings being 4 out of 5, and all users stating their interest in using MoneyBrew.

Table 5. Overall application evaluation descriptive statistics results.

Overall app evaluation descriptive statistics (<i>n</i> = 10)	
User ratings (Scale 1 – 5)	
Visual appearance, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	7 (70)
Score 5	2 (20)
Mode	4
Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	4.00
Quartile 3 (75 th percentile)	4.25
Overall experience, <i>n</i> (%)	
Score 1	0 (0)
Score 2	0 (0)
Score 3	1 (10)
Score 4	5 (50)
Score 5	4 (40)
Mode	4
Quartile 1 (25 th percentile)	4.00
Quartile 2 (50 th percentile, median)	4.00
Quartile 3 (75 th percentile)	5.00
Would use the app, <i>n</i> (%)	
Yes	10 (100)
No	0 (0)

Statistical inference analysis

All task-specific metrics were analyzed in conjunction with a descriptive grouping variable: age group (according to the groups specified in the sample characterization table), highest completed education level (hereinafter referred to as education level), experience with mobile technologies (mobile experience), and prior usage of similar apps (similar app usage). Although we intended to perform equivalent tests with the employment status variable as a grouping factor, given that only one group included more than one individual, statistical testing

could not be carried out. Moreover, due to a small occurrence of misclick events, statistical analyses considering this as a dependent variable were also not conducted.

For each comparison, and when applicable (numerical dependent variable) approximation to normality was determined using the Shapiro-Wilk test and, when the tests were not statistically significant, the homogeneity of variances assumption was verified using a Levene's test. In particular, if the Shapiro-Wilk test was determined to be significant for at least one of the groups in testing, the overall result was deemed to be significant. When approximation to normality and homogeneity of variances were verified (both tests rendering non-significant), a parametric test was conducted, otherwise a non-parametric alternative was selected. However, if one of the groups of the independent variable was only comprised of two elements, a non-parametric test was immediately selected, with no normality nor variances testing performed. We considered a significance level, α , of 0.05 on all performed statistic tests. Given that statistical testing could not be performed on all intended comparisons, graphical representations were only presented to illustrate those in which we did so.

Importantly, classification scores correspond to categorical and ordinal variables. As such, we intended to use Chi-square tests to compare the various collected ratings and each of the four selected independent variables; however, this test implies an assumption that all individual cells of the associated contingency table include more than five data points. Since that was not possible, considering our small sample size, and because the alternative Fisher's exact test can only be performed for 2×2 contingency tables, we considered these dependent variables to be continuous variables, and performed the corresponding statistical tests.

The performed tests ended up being of four possible types: the parametric one-way independent samples t -test or ANOVA (analysis of variance) (independent variable comprised of two or more groups, respectively) or, conversely, the non-parametric Mann-Whitney U or Kruskal-Wallis test. All results that could be obtained for each collected data category are presented below.

Task 1: Ask the AI chatbot, Brewy, to explain what the 50-30-20 financial model is

Considering our overall strategy previously detailed, statistical testing on the data related to the first of the three tasks was only performed on three variables: time to complete the task itself, classification of the speed taken to execute the task, and the visual appearance score of the app on the visited screens. For the three variables, no statistical differences were found, with all p -values being above the significance threshold. All task 1 results are presented on Table 6, with graphical representations of the comparisons in Figure 1.

Table 6. Task 1 statistical inference results. N/A: not assessed.

Task 1 statistical analyses					
Dependent variable	Independent variable	Normality testing	Homogeneity of variances testing	Statistical test	p-value
Full task completion	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Task completion time	Age group	Non-significant	Non-significant	One-way ANOVA	0.807
	Education level	Non-significant	Non-significant	One-way ANOVA	0.947
	Mobile experience	Non-significant	Non-significant	One-way ANOVA	0.539
	Similar app usage	Non-significant	Non-significant	Independent samples <i>t</i> -test	0.918
Efficiency rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Speed rating	Age group	Significant	N/A	Kruskal-Wallis test	1.000
	Education level	Significant	N/A	Kruskal-Wallis test	0.491
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.641
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.413
Ease of execution rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Visual appearance rating	Age group	Significant	N/A	Kruskal-Wallis test	0.260
	Education level	Significant	N/A	Kruskal-Wallis test	0.114
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.906
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.190
Overall experience rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Would use task's functionality	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A

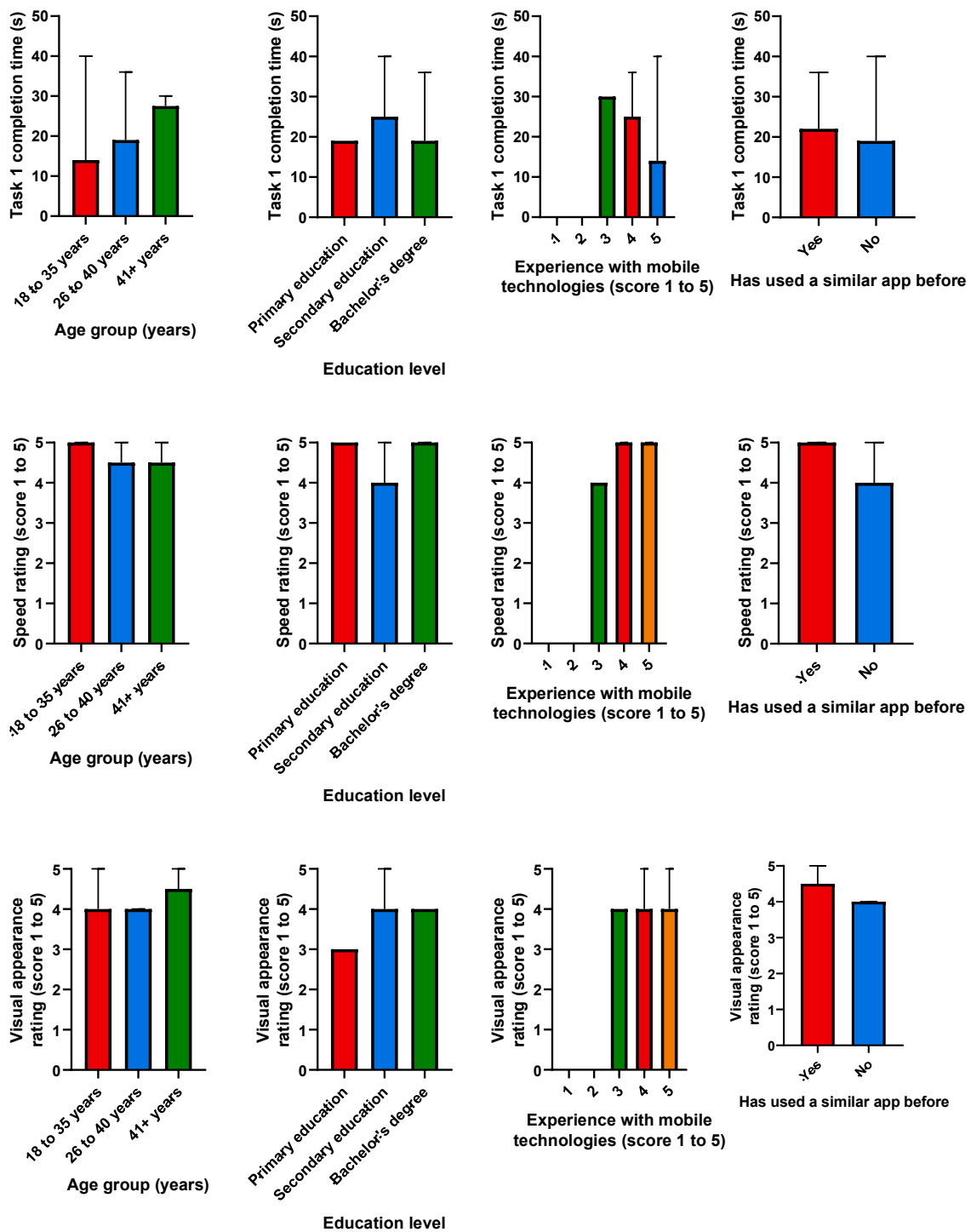


Figure 1. Task 1 graphical representations of analyzed statistical comparisons. Plots represent median value with 95% confidence interval.

Task 2: Remove an inactive Spotify subscription after receiving an automatic notification informing the user that they haven't listened to music for over 10 days

Obtained results for the smart notification task, presented on Table 7 and represented on Figure 2 plots, were carried out on the time to complete and visual appearance rating data. Similar to the first task, no significant differences were detected between the different factoring groups.

Table 7. Task 2 statistical inference results. N/A: not assessed.

Task 2 statistical analyses					
Dependent variable	Independent variable	Normality testing	Homogeneity of variances testing	Statistical test	p-value
Full task completion	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Task completion time	Age group	Non-significant	Non-significant	One-way ANOVA	0.191
	Education level	Non-significant	Non-significant	One-way ANOVA	0.498
	Mobile experience	Non-significant	Non-significant	One-way ANOVA	0.221
	Similar app usage	Non-significant	Non-significant	Independent samples <i>t</i> -test	0.174
Efficiency rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Speed rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Ease of execution rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Visual appearance rating	Age group	Significant	N/A	Kruskal-Wallis test	0.368
	Education level	Significant	N/A	Kruskal-Wallis test	0.707
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.704
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.257
Overall experience rating	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Would use task's functionality	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A

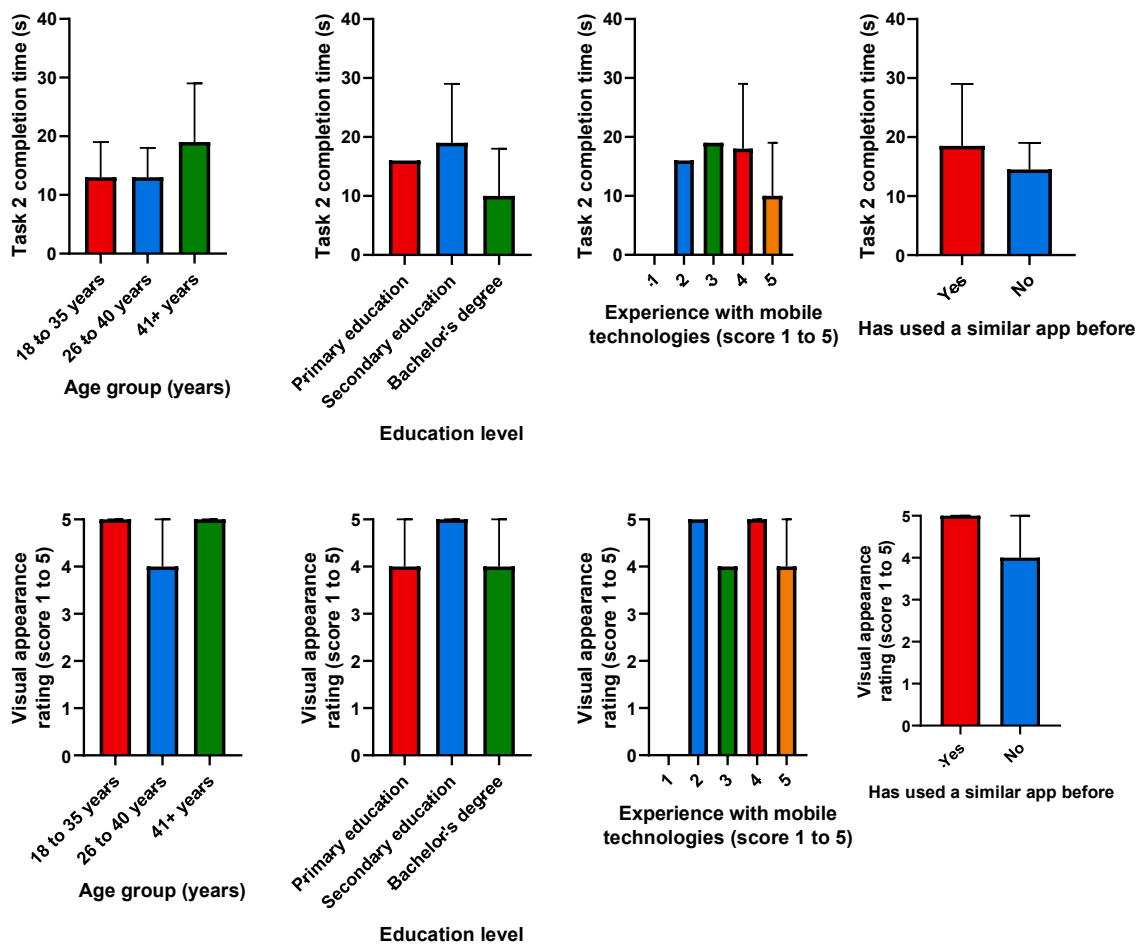


Figure 2. Task 2 graphical representations of analyzed statistical comparisons. Plots represent median value with 95% confidence interval.

Task 3: Access a specific learning page (with video and article) about Work Contracts, and take a small quiz about the learned content

For the last task, and because results were not as uniform as the previous two tasks, we were able to conduct tests on the completion time data and on all classification score metrics. In these results, we were, indeed, able to obtain a statistically significant result for the visual appearance rating, when comparing individuals by their previous usage of an app similar to MoneyBrew ($p = 0.038$, Table 8). As can be visually observed in the corresponding plot, presented in Figure 3, individuals who had previously used a financial management app provided higher scores than those who had no prior contact with such an app.

Table 8. Task 3 statistical inference results. N/A: not assessed.

Task 3 statistical analyses					
Dependent variable	Independent variable	Normality testing	Homogeneity of variances testing	Statistical test	p-value
Full task completion	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A
Task completion time	Age group	Non-significant	Non-significant	One-way ANOVA	0.452
	Education level	Non-significant	Non-significant	One-way ANOVA	0.630
	Mobile experience	Non-significant	Non-significant	One-way ANOVA	0.405
	Similar app usage	Non-significant	Non-significant	Independent samples t-test	0.100
Efficiency rating	Age group	Significant	N/A	Kruskal-Wallis test	0.552
	Education level	Significant	N/A	Kruskal-Wallis test	0.939
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.572
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.610
Speed rating	Age group	Significant	N/A	Kruskal-Wallis test	0.174
	Education level	Significant	N/A	Kruskal-Wallis test	0.377
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.256
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.610
Ease of execution rating	Age group	Significant	N/A	Kruskal-Wallis test	0.472
	Education level	Significant	N/A	Kruskal-Wallis test	0.430
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.212
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.476
Visual appearance rating	Age group	Significant	N/A	Kruskal-Wallis test	0.458
	Education level	Significant	N/A	Kruskal-Wallis test	0.920
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.739
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.038
Overall experience rating	Age group	Significant	N/A	Kruskal-Wallis test	0.645
	Education level	Significant	N/A	Kruskal-Wallis test	0.747
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.137
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.114
Would use task's functionality	Age group	N/A	N/A	N/A	N/A
	Education level	N/A	N/A	N/A	N/A
	Mobile experience	N/A	N/A	N/A	N/A
	Similar app usage	N/A	N/A	N/A	N/A

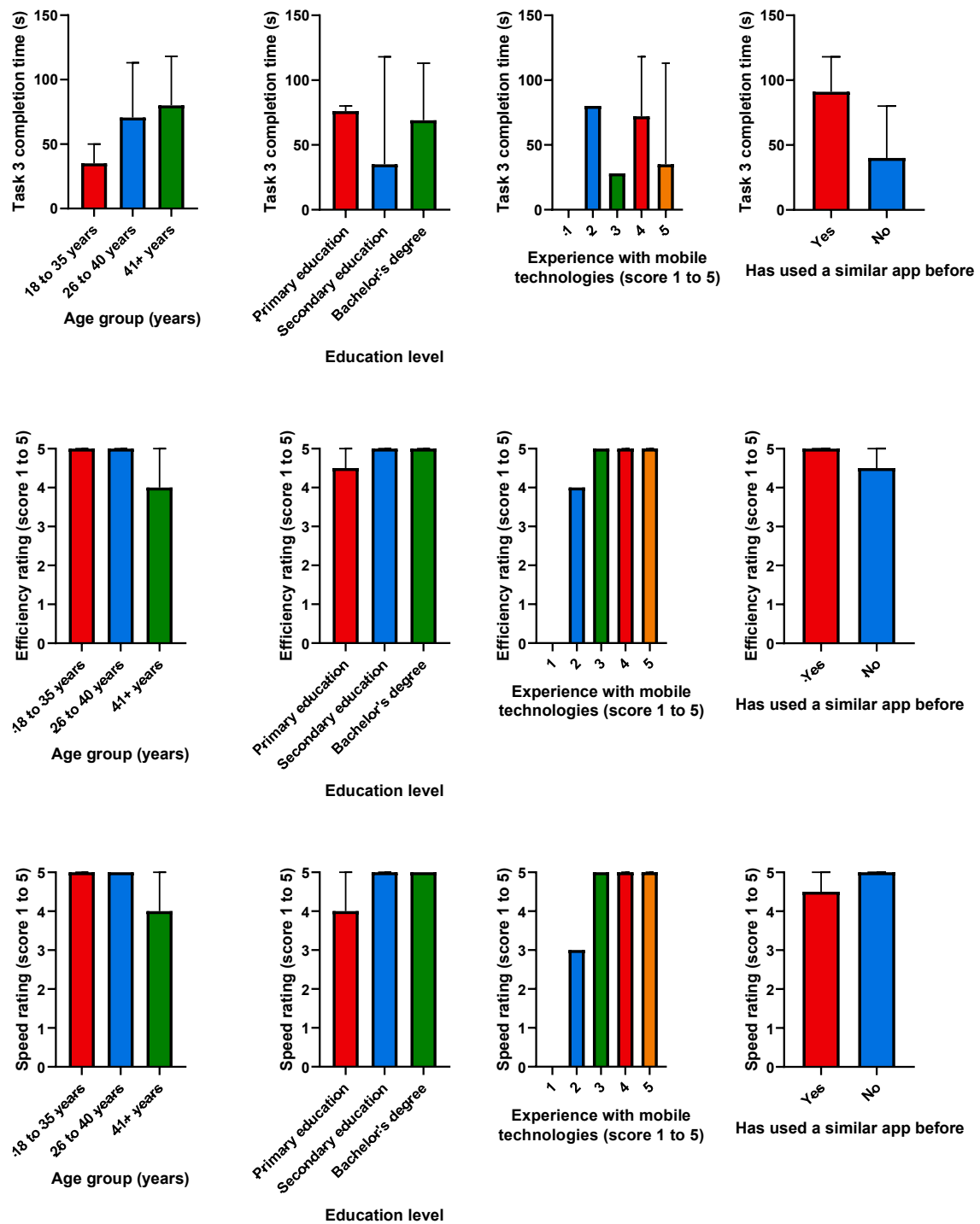


Figure 3.1. Task 3 graphical representations of analyzed statistical comparisons. Plots represent median value with 95% confidence interval.

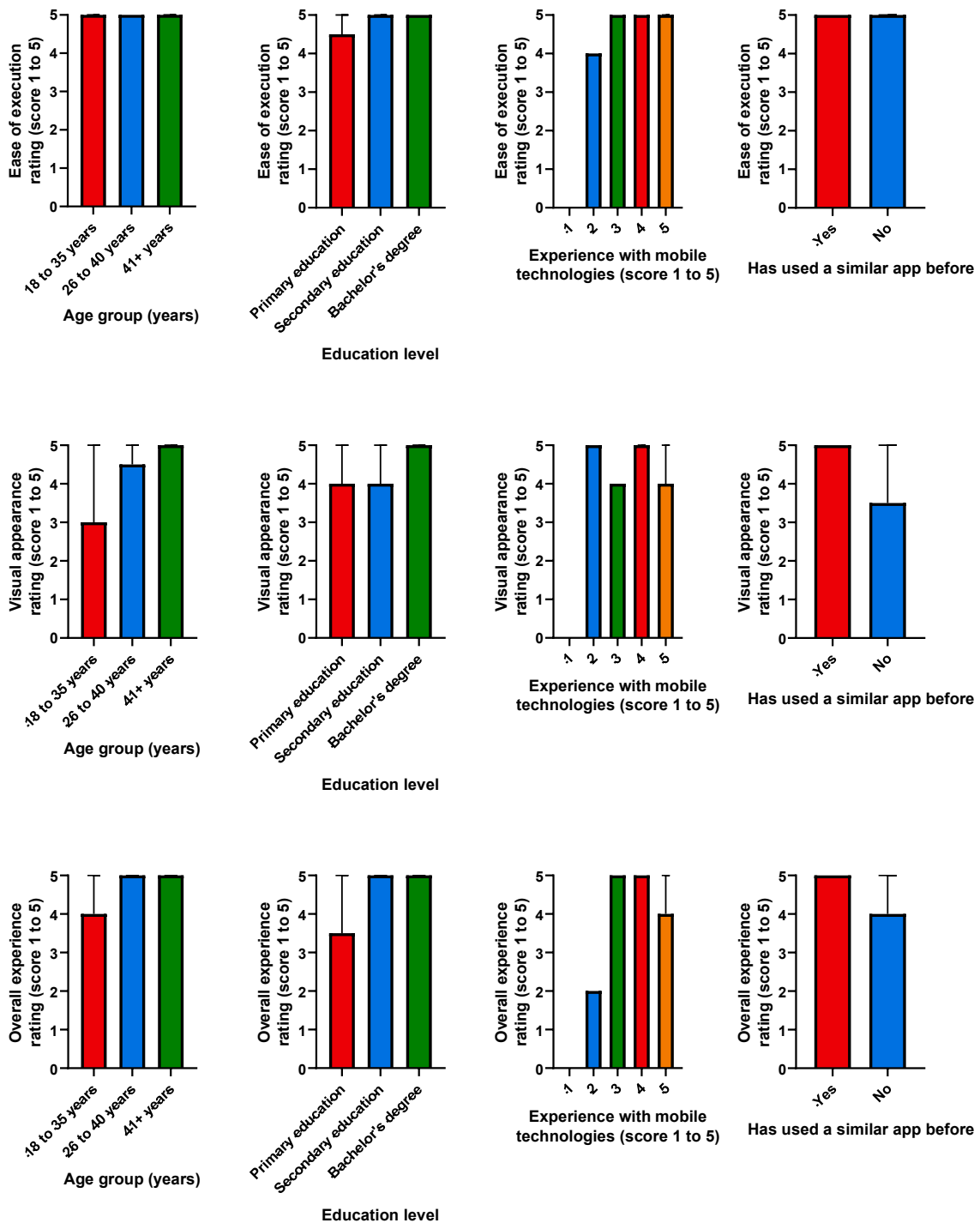


Figure 3.2. Task 3 graphical representations of analyzed statistical comparisons. Plots represent median value with 95% confidence interval.

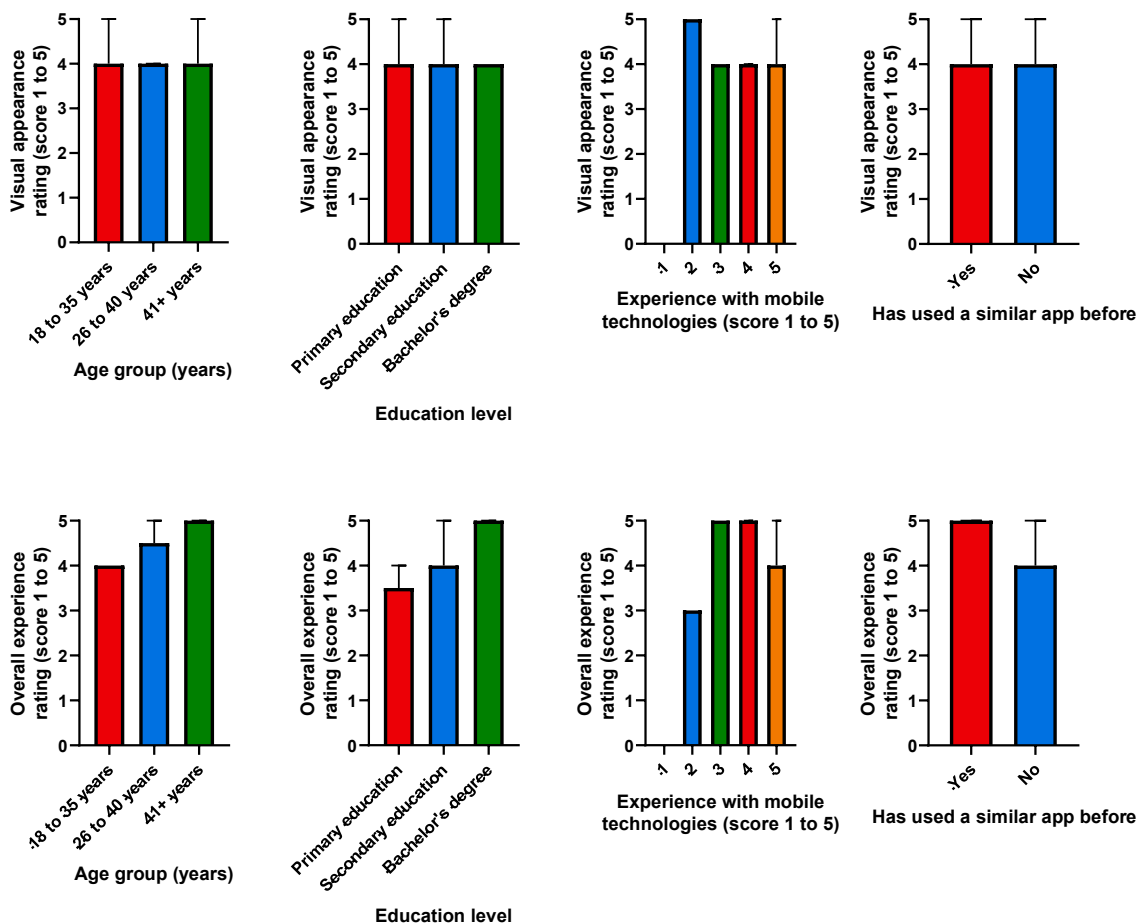
Overall application evaluation

As for the overall app evaluation, statistical testing was conducted on both collected scores (as all individuals claimed to have interest in using the app, no testing was carried out regarding this variable). However, in accordance with most of the task-specific tests, no significant results were obtained (Table 9, Figure 4).

Table 9. General application evaluation statistical inference results. N/A: not assessed.

General application evaluation statistical analyses					
Dependent variable	Independent variable	Normality testing	Homogeneity of variances testing	Statistical test	p-value
Visual appearance rating	Age group	Significant	N/A	Kruskal-Wallis test	0.287
	Education level	Significant	N/A	Kruskal-Wallis test	0.870
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.210
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.610
Overall experience rating	Age group	Significant	N/A	Kruskal-Wallis test	0.522
	Education level	Significant	N/A	Kruskal-Wallis test	0.186
	Mobile experience	Significant	N/A	Kruskal-Wallis test	0.169
	Similar app usage	Significant	N/A	Mann-Whitney U test	0.114

Figure 4. graphical representations of analyzed statistical comparisons regarding overall app evaluation. Plots represent median value with 95% confidence interval.



Discussion

The evaluation of MoneyBrew's prototype by the users turned out to be generally good among the three tested functionalities, as the participants revealed ease in successfully completing all tasks and gave overall favorable judgments of the interface. These results suggest that the design interventions applied in Phase III, especially in what concerns the major usability problems identified during the heuristic evaluation, were efficient in bettering the user's global experience.

Completion rates in all three tasks exceeded the previously established success thresholds. In Task 1, we were able to achieve a 90% completion rate, as 9 in 10 users successfully interacted with Brewy and received a detailed explanation of the financial model 50-30-20, and above the 80% goal. The mean completion time was 22.9 seconds, which remained well within the 20-second efficiency target, suggesting that users were able to easily locate and access the chatbot functionality.

Task 2 accomplished a perfect completion rate, with all 10 participants successfully removing an inactive Spotify subscription as recommended by the app. This represents a strong improvement compared to the initial concern raised in the heuristic evaluation related to the visual prominence of the cancelation button (that had been identified as a possible inducer of unintentional actions). The implementation of a balanced visual hierarchy between the buttons "Cancel subscription" and "Keep active" seems to have fixed the problem. The mean time of 15.50 seconds shows that the simplified interface allowed users to make informed decisions about the subscription management with as little friction as possible.

Task 3, the most complex out of all tasks, was successfully completed (either fully or partially) by all participants, even though 2 users did not complete the quiz rating step. The inclusion of these data in the analysis remains justified, since the unavailability of the rating functionality does not jeopardize the main goal of accessing and interacting with the financial literacy content, even more so given the reason why these users failed to fulfill this last step was due to a misunderstanding of their need to do so. This task's mean time of completion was 61.04 seconds, which reflects the added difficulty of navigating the hub, consuming educational content and completing the quiz. This value was well below the original estimation of 5 to 10 minutes, however, this estimation was largely affected by the inherent fluctuation caused by the level of difficulty of each question presented.

Overall, participants reported a positive experience when interacting with the MoneyBrew prototype. Across the three evaluated tasks, the ratings for efficiency, speed, and ease of execution were consistently high, indicating that users generally completed the tasks quickly and with minimal effort. In fact, the first two tasks were particularly well received, with median scores of 5 out of 5 across most usability dimensions. Although the third task showed slightly more variation (likely due to its greater complexity), it still achieved high median ratings, indicating that users remained comfortable navigating the interface even when engaging with more demanding content.

Visual appearance was the only aspect that consistently received slightly lower ratings. Across the three tasks, median scores for this dimension ranged between 4 and 4.5 out of 5, a tendency that was also reflected in the overall application evaluation, where both visual appearance and overall experience were rated at a median of 4 out of 5. This suggests that,

although the interface supported task completion effectively and did not generate confusion, users perceived the visual design as somewhat less strong than other elements of the experience. Considering that simplicity had previously been identified as an important user preference, these results lead us to conclude that we may have placed too great an emphasis on clarity and functionality and ended up lacking in visual expressiveness.

Statistical analysis revealed a high level of consistency across demographic groups. For the first two tasks, no significant differences were found in task performance or user ratings when comparing participants by age, education level, mobile experience, or prior use of similar applications. This lack of variation suggests that the interface is accessible and intuitive for a broad range of users, which supports its suitability for the intended target audience.

A single statistically significant result emerged in the third task, where visual appearance ratings differed based on prior experience with financial management applications ($p = 0.038$). Participants with previous exposure to similar tools rated the visual design more positively than those without such experience. This finding may indicate that experienced users more readily recognized familiar design patterns, or that the prototype aligns well with established conventions in financial applications.

All ten participants expressed a clear willingness to use MoneyBrew if it were officially launched. This unanimous interest is a strong indicator of the app's perceived value, suggesting that the core features genuinely resonate with the target audience and that the interaction flows feel both clear and trustworthy. While the overall experience was rated positively, the visual appearance consistently received slightly lower marks compared to functional performance. This suggests that while the interface supports task success effectively, the aesthetic side may need to look more polished to sustain long-term engagement for daily use. However, these results must be viewed in light of several study limitations. First, the evaluation was based on a small sample of 10 individuals. Because this was a convenience sample that included friends and family, there is a distinct possibility of social desirability bias. Essentially, participants may have been more patient or kinder in their feedback than an objective stranger would be. Additionally, the sample was largely composed of "mobile-tech-savvy" individuals, with 80% of users rating their tech experience at a 4 or 5 out of 5. This means the results may not fully represent the challenges that less experienced users might encounter.

Despite these factors, the evaluation provides strong evidence that MoneyBrew's core design is effective. Every participant expressed a genuine interest in using the app if it were officially released, confirming that the tool addresses a real demand for personal financial control. The fact that users could consistently complete tasks with minimal difficulty suggests that the interface is fundamentally intuitive and well-aligned with user expectations.

Conclusion

The conception and development of MoneyBrew was driven by a realistic need, since individuals find financial management challenging and time-consuming, and existing solutions have failed to achieve widespread adoption. Following the project requirements, we executed an iterative design process, beginning with robust user research, and then refining the interface based on both peer critique and user testing.

In Phase I, we conducted a questionnaire with 38 participants to understand their financial management practices and expectations. While the vast majority expressed interest in digital financial tools, the actual usage remained surprisingly low, with only a quarter of respondents actively using such applications. This disconnect pointed to severe issues, either in usability or design, in existing solutions. More importantly, the research revealed consistent user priorities: simplicity and privacy were fundamental, followed by specific feature requests around expense automation and financial education. These findings shaped our design decisions throughout the project.

Then, in Phase II, we developed a low-fidelity prototype centered on three functionalities that directly addressed user needs: an AI assistant for financial guidance, intelligent notifications for subscription management, and a financial literacy hub. When colleagues evaluated this prototype using heuristic methods, they identified several meaningful issues, namely the button prominence that could inadvertently bias user behavior, and vague messaging in the chatbot interface. Rather than dismissing this feedback, we treated it as essential guidance for improvement.

The refinements made in Phase III yielded measurable results. When the updated prototype was tested by 10 participants, we found strong performance across all tasks. Task completion exceeded our thresholds and, more tellingly, users rated their experience highly in terms of efficiency and ease. The one statistically significant finding, that stated that users with prior experience using financial apps rated the visual design more favorably, suggests our interface aligns well with established conventions in the field. But, undoubtedly, the most validating part was the unanimous response to whether users would adopt the application: all ten participants said yes.

This work demonstrates that iterative design grounded in user research produces more effective outcomes than design done in isolation. The path from initial concept through peer critique and to user validation revealed both strengths and areas for refinement. While the visual design could be enhanced further, and our sample size carries inherent limitations, the consistent positive feedback and successful task completion suggest we achieved our core objective of creating a financial management tool that is usable and addresses real user needs.

Although MoneyBrew remains a fictional application prototype without commercial intent, the process has illustrated fundamental principles of human-computer interaction, including the value of understanding users before designing, the importance of iterative refinement based on feedback, and the necessity of empirical testing to validate design decisions. In this sense, the project succeeds, not as a product, but as a learning experience in effective user-centered design methodology.

Annexes

Annex I: Complete questionnaire

Below is the complete questionnaire we implemented in this first phase of our project. It was developed in Portuguese, given that all respondents speak this language. Circular radio buttons are presented in questions in which participants could only select one of the provided options, whereas squared checkboxes are represented when the respondent could select more than one option. Of note, the section on financial investing was presented only to those who indicated that they currently have financial investments.

Questionário | Hábitos de gestão financeira

Olá! Somos um grupo de estudantes de 3.º ano da Licenciatura em Engenharia Informática e Computação da Faculdade de Engenharia da Universidade do Porto, e estamos a desenvolver um protótipo de uma aplicação de gestão financeira, no âmbito da unidade curricular de Interação Pessoa-Computador. Nesse sentido, desenvolvemos este questionário, de forma a conhecermos melhor os hábitos dos potenciais utilizadores da nossa aplicação.

Respostas a este questionário são anónimas, e os dados recolhidos utilizados exclusivamente para o fim apresentado. Não há respostas certas ou erradas, apenas solicitamos que leia atentamente cada pergunta e responda com a maior honestidade possível.

Agradecemos, desde já, o tempo dispensado! 😊

Informações demográficas

Qual a sua faixa etária?

- ☐ Menor de idade (menos de 18 anos)
- ☐ Entre 18 e 25 anos
- ☐ Entre 26 e 35 anos
- ☐ Entre 36 e 45 anos
- ☐ Entre 46 e 55 anos
- ☐ Entre 56 e 65 anos
- ☐ Mais de 65 anos

Qual o seu estado civil?

- ☐ Solteiro(a)
- ☐ Casado(a)
- ☐ União de facto
- ☐ Divorciado(a)
- ☐ Viúvo(a)

Qual o seu nível de escolaridade?

Selecione, por favor, a opção onde se enquadra o nível mais elevado que possui.

- ☐ Ensino básico (1.º, 2.º ou 3.º ciclos)
- ☐ Ensino secundário: cursos científico-humanísticos (Ciências e Tecnologias, Línguas e Humanidades, Ciências Socioeconómicas, ou equivalentes)
- ☐ Ensino secundário: cursos profissionais
- ☐ Ensino superior: licenciatura
- ☐ Ensino superior: mestrado
- ☐ Ensino superior: doutoramento

Qual a sua situação profissional atual?

- ☐ Estudante [não-trabalhador(a)]
- ☐ Trabalhador(a)-estudante
- ☐ Trabalhador(a) por conta de outrem
- ☐ Trabalhador(a) por conta própria
- ☐ Desempregado(a)
- ☐ Reformado(a)
- ☐ Outra: _____

Situação financeira atual, e gastos e ganhos habituais

Por quantas pessoas é constituído o seu agregado familiar?

- ☐ 1 [vivo sozinho(a)]
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6 ou mais

Por quantos dependentes é constituído o seu agregado familiar?

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3 ou mais

De forma aproximada, qual é o rendimento mensal do seu agregado familiar?

- ☐ Menos de 500 €
- ☐ Entre 500 e 749 €
- ☐ Entre 750 e 999 €
- ☐ Entre 1000 e 1499 €
- ☐ Entre 1500 e 1999 €
- ☐ Entre 2000 e 2999 €
- ☐ Entre 3000 e 4999 €
- ☐ Entre 5000 e 7499 €
- ☐ Igual ou superior a 7500 €
- ☐ Não sei

☐ Prefiro não responder

Quais os gastos mensais habituais do seu agregado familiar que são recorrentes e inadiáveis (gastos que tem todos os meses e dos quais não pode abdicar)?

- ☐ Renda da casa
- ☐ Empréstimo bancário
- ☐ Seguro
- ☐ Impostos
- ☐ Saúde
- ☐ Gás e eletricidade
- ☐ Água
- ☐ Combustível
- ☐ Telemóvel
- ☐ Telefone fixo
- ☐ Internet
- ☐ Canais de televisão
- ☐ Escola e/ou universidade
- ☐ Transportes públicos
- ☐ Subscrições de entretenimento (por exemplo, Netflix, HBO Max, Disney+, SportTV, etc.)
- ☐ Atividades de exercício físico
- ☐ Outra: _____

Quais os gastos mensais habituais do seu agregado familiar que não são essenciais, e que poderia, caso fosse necessário, descartar a curto ou longo prazo?

- ☐ Renda da casa
- ☐ Empréstimo bancário
- ☐ Seguro

- ☐ Impostos
- ☐ Saúde
- ☐ Gás e eletricidade
- ☐ Água
- ☐ Combustível
- ☐ Telemóvel
- ☐ Telefone fixo
- ☐ Internet
- ☐ Canais de televisão
- ☐ Escola e/ou universidade
- ☐ Transportes públicos
- ☐ Subscrições de entretenimento (por exemplo, Netflix, HBO Max, Disney+, SportTV, etc.)
- ☐ Atividades de exercício físico
- ☐ Outra: _____

Atualmente, possui algum tipo de investimentos financeiros?

- ☐ Sim
- ☐ Não, mas gostaria de o fazer no futuro
- ☐ Não, e não tenho intenção em fazê-lo

Investimentos financeiros

(Aplicado apenas a quem respondeu que possui investimentos financeiros na questão anterior)

Quais os tipos de investimentos financeiros que possui?

- ☐ Contas-poupança
- ☐ Imóveis

- ☐ Ações
- ☐ ETFs
- ☐ Criptomoedas
- ☐ Obrigações
- ☐ Matérias-primas
- ☐ Outra: _____

Como classifica o seu grau de envolvimento nos seus investimentos financeiros?

- ☐ Não tenho conhecimento na área, pelo que invisto apenas após recomendação por parte de outra pessoa ou entidade
- ☐ Invisto de forma mais instintiva, pois possuo um conhecimento mais limitado na área
- ☐ Sigo ativamente os mercados e tendências, adequando os meus investimentos aos mesmos
- ☐ Sigo ativamente os mercados e tendências, adequando os meus investimentos aos mesmos
- ☐ Não lido pessoalmente com os meus investimentos, recorrendo a serviços profissionais que o façam por mim
- ☐ Outra: _____

Quais os seus objetivos ao fazer investimentos?

- ☐ Ganho monetário imediato
- ☐ Combate à inflação
- ☐ Poupança para a reforma
- ☐ Outra: _____

Hábitos atuais de gestão financeira

Com que frequência controla as suas despesas?

- ☐ Diariamente (pondero todas as minhas despesas de acordo com o meu orçamento)
- ☐ Semanalmente
- ☐ Mensalmente
- ☐ Raramente (apenas para despesas mais avultadas)
- ☐ Nunca (se preciso ou quero algo, então compro, assumindo que tenho orçamento para tal)
- ☐ Nunca (outra pessoa do meu agregado familiar é responsável por essa tarefa)

No último ano, e de forma aproximada, com que frequência cumpriu o seu orçamento mensal?

- ☐ Sempre (todos os meses)
- ☐ Entre 8 e 11 meses
- ☐ Entre 4 e 7 meses
- ☐ Entre 1 e 3 meses
- ☐ Nunca (fui além do meu orçamento em todos os meses)
- ☐ Não faço orçamentos mensais/não sou responsável por essa tarefa

No último ano, e de forma aproximada, com que frequência conseguiu poupar?

- ☐ Sempre (todos os meses)
- ☐ Entre 8 e 11 meses
- ☐ Entre 4 e 7 meses
- ☐ Entre 1 e 3 meses
- ☐ Nunca
- ☐ Não sou responsável por essa tarefa

Quais dos seguintes tipos de registo de despesas já utilizou?

- ☐ Caderno ou bloco de notas (formato físico)
- ☐ Bloco de notas digital (telemóvel, computador ou tablet)
- ☐ Folha de cálculo (por exemplo, Excel)
- ☐ Aplicação móvel desenvolvida para esse efeito
- ☐ Nunca registei as minhas despesas
- ☐ Outra: _____

Dos seguintes desafios, quais considera serem os que mais o(a) impedem de controlar a sua gestão financeira?

- ☐ Falta de registo de despesas
- ☐ Dificuldade em poupar, tendo em conta os meus gastos habituais
- ☐ Incapacidade de prever despesas habituais
- ☐ Surgimento de despesas de última hora
- ☐ Gastos impulsivos
- ☐ Falta de motivação ou tempo para organizar as minhas finanças
- ☐ Outra: _____

Utilidade de uma nova solução para a sua gestão financeira

Com que frequência utiliza uma aplicação de gestão financeira?

- ☐ Diariamente
- ☐ Semanalmente
- ☐ Mensalmente
- ☐ Anualmente
- ☐ Já utilizei, mas não o faço atualmente
- ☐ Nunca utilizei

O que o(a) interessa neste tipo de aplicações?

Selecione, por favor, até 3 interesses que considera mais relevantes.

- ☐ Simplicidade de utilização
- ☐ Intuitiva e visualmente apelativa
- ☐ Segurança e privacidade
- ☐ Integração com serviços externos (por exemplo, contas bancárias)
- ☐ Acompanhamento personalizado
- ☐ Outra: _____

Quais das seguintes funcionalidades considera mais interessantes nas aplicações de gestão financeira que existem hoje em dia?

- ☐ Registo automático de despesas
- ☐ Análise de padrões habituais de consumo e investimento
- ☐ Relatórios inteligentes de evolução da sua situação financeira
- ☐ Orçamento e controlo de gastos
- ☐ Definição e acompanhamento de objetivos (por exemplo, fundo de emergência)
- ☐ Agregação de contas financeiras (por exemplo, bancos, corretoras, serviços, etc.)
- ☐ Colaboração e contas partilhadas
- ☐ Outra: _____

Quais das seguintes inovações considera interessantes numa potencial nova aplicação de gestão financeira?

- ☐ Chatbot (assistente virtual) de apoio à decisão, com recurso a inteligência artificial
- ☐ Formação na área de literacia financeira
- ☐ Gestão de despesas diretamente na aplicação (por exemplo, subscrição e cancelamento de serviços)
- ☐ Análise pormenorizada dos hábitos de consumo (por exemplo, identificar a frequência de compra de um produto num determinado serviço)
- ☐ Sugestão de modelos de gestão financeira a adotar (por exemplo, modelo 50-30-20)

Tem mais alguma sugestão de funcionalidade inovadora que gostaria que uma aplicação deste género incluísse?

Annex II: Summary of results

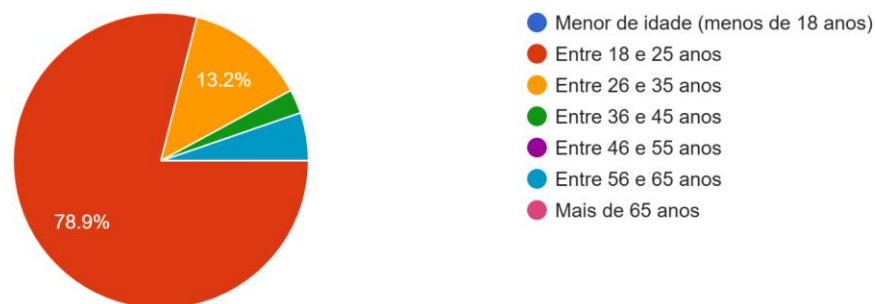
The findings reveal a young, educated, and technologically aware population that shows growing concern for financial control, yet struggles with consistency in saving and long-term planning. The high percentage of participants expressing interest in investment but lacking current engagement suggests a gap in financial literacy and accessible tools.

Furthermore, the limited adoption of financial management apps underscores the need for solutions that are both intuitive and motivating, combining simplicity, personalized insights, and educational support to foster sustainable financial habits.

Below are the graphical representations of the answers to all 23 of the questions included in our survey, automatically generated by the Google Forms platform.

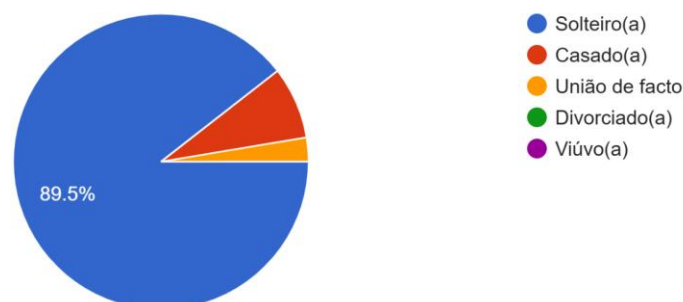
Qual a sua faixa etária?

38 responses



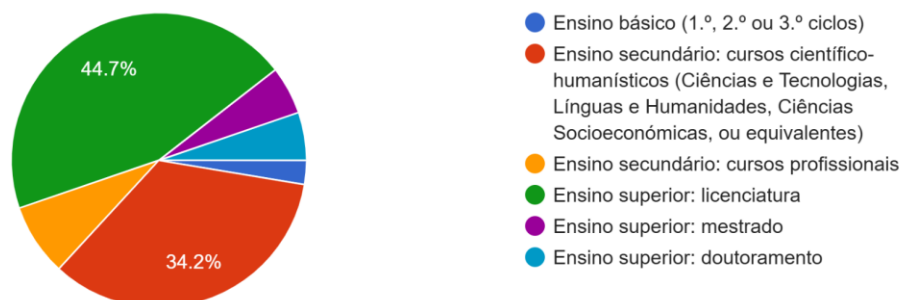
Qual o seu estado civil?

38 responses



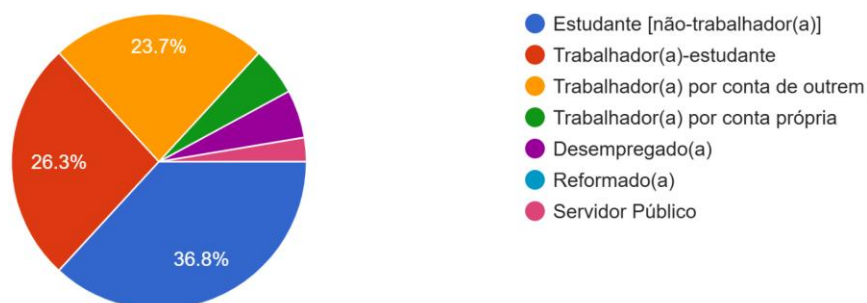
Qual o seu nível de escolaridade? Selecione, por favor, a opção onde se enquadra o nível mais elevado que possui.

38 responses



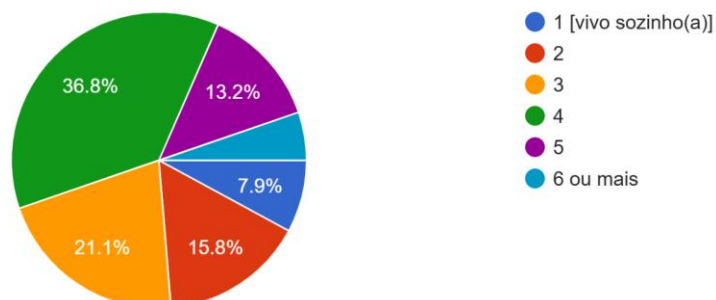
Qual a sua situação profissional atual?

38 responses



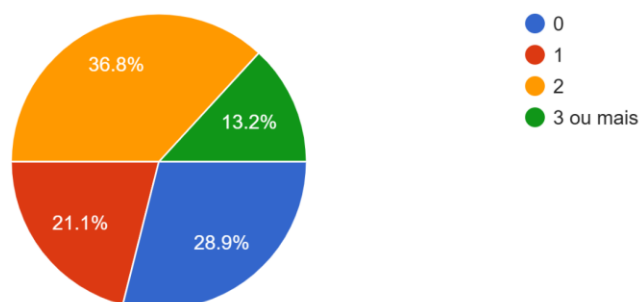
Por quantas pessoas é constituído o seu agregado familiar?

38 responses



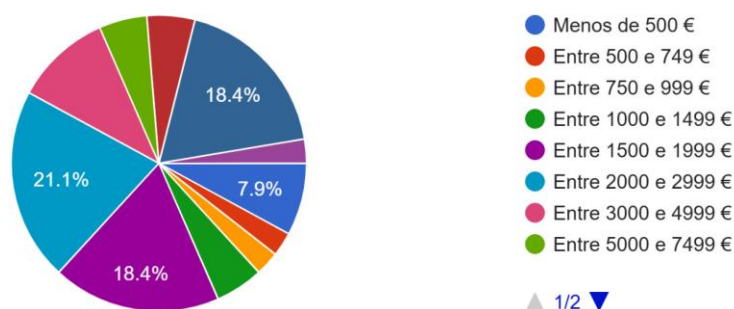
Por quantos dependentes é constituído o seu agregado familiar?

38 responses



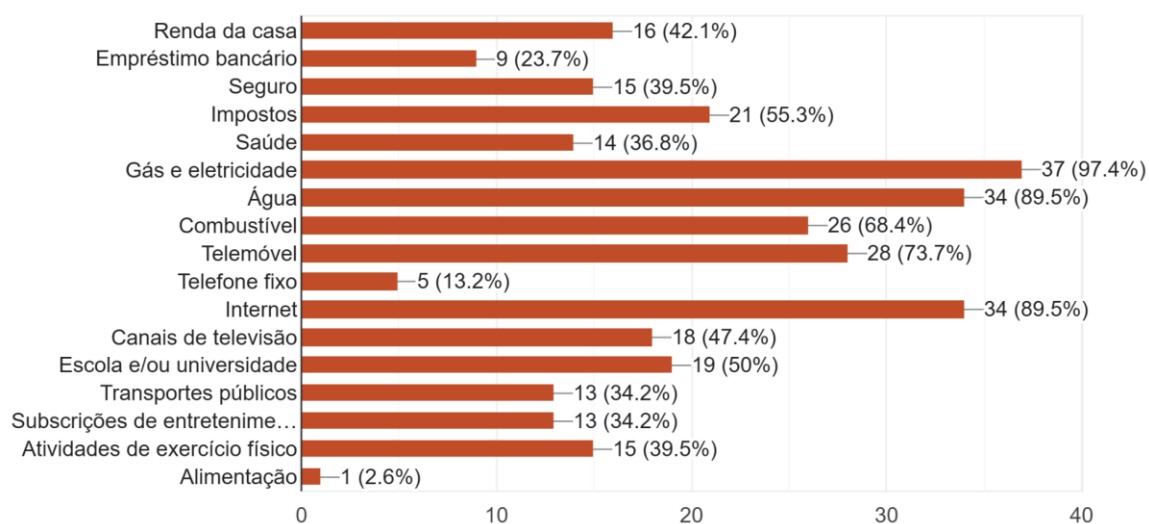
De forma aproximada, qual é o rendimento mensal do seu agregado familiar?

38 responses



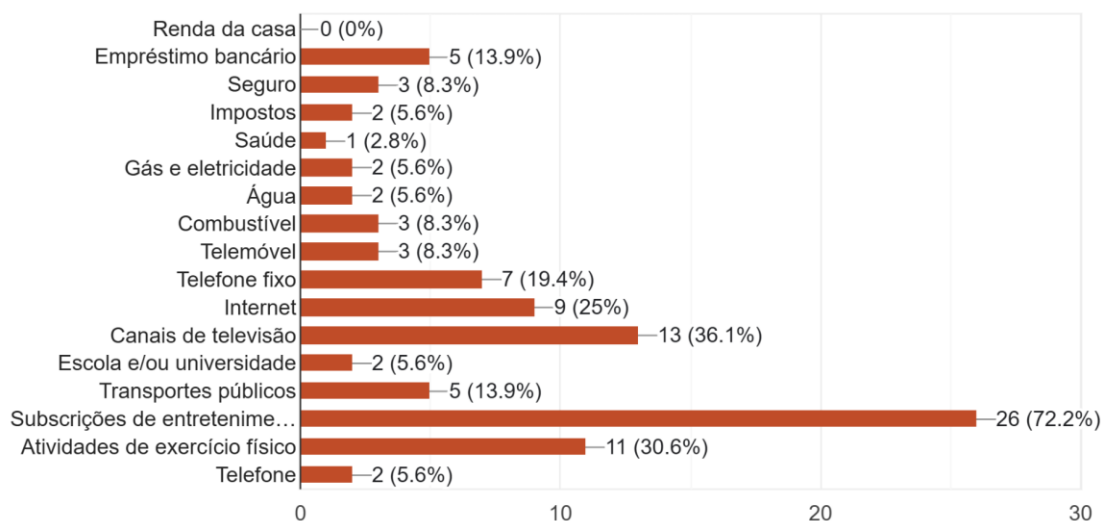
Quais os gastos mensais habituais do seu agregado familiar que são recorrentes e inadiáveis (gastos que tem todos os meses e dos quais não pode abdicar)?

38 responses



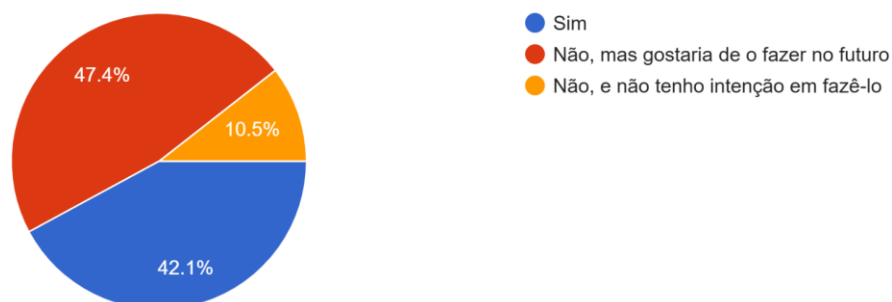
Quais os gastos mensais habituais do seu agregado familiar que não são essenciais, e que poderia, caso fosse necessário, descartar a curto ou longo prazo?

36 respostas



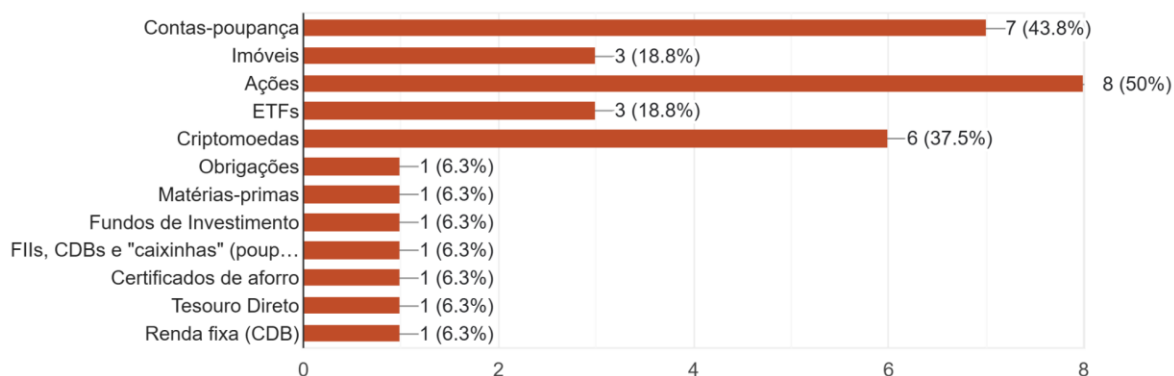
Atualmente, possui algum tipo de investimentos financeiros?

38 respostas



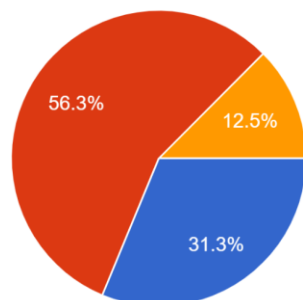
Quais os tipos de investimentos financeiros que possui?

16 respostas



Como classifica o seu grau de envolvimento nos seus investimentos financeiros?

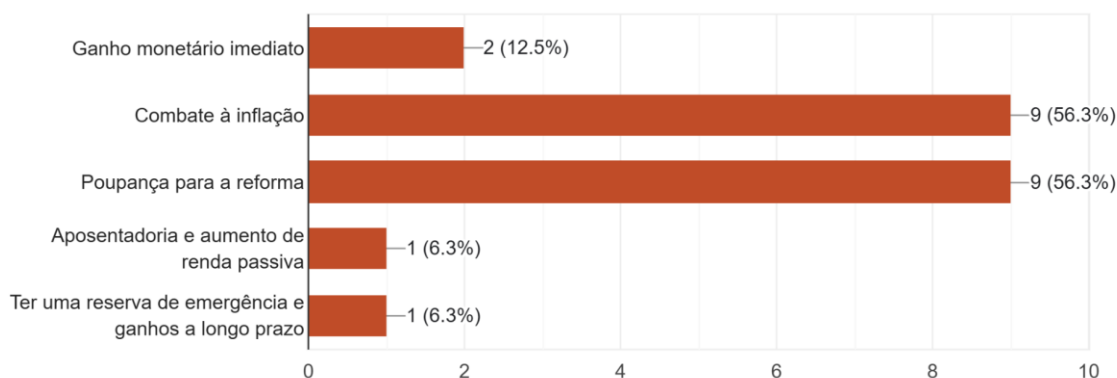
16 responses



- Não tenho conhecimento na área, pelo que invisto apenas após recomendação por parte de outra pessoa ou entidade
- Invisto de forma mais instintiva, pois possuo um conhecimento mais limitado na área
- Sigo ativamente os mercados e tendências, adequando os meus investimentos
- Não lido pessoalmente com os meus investimentos, recorrendo a serviços...

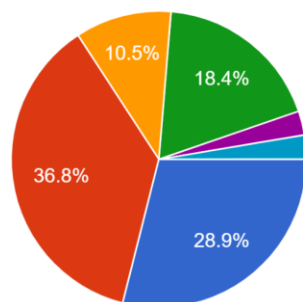
Quais os seus objetivos ao fazer investimentos?

16 responses



Com que frequência controla as suas despesas?

38 responses



- Diariamente (pondero todas as minhas despesas de acordo com o meu orçamento)
- Semanalmente
- Mensalmente
- Raramente (apenas para despesas mais avultadas)
- Nunca (se preciso ou quero algo, então compro, assumindo que tenho orçamento)
- Nunca (outra pessoa do meu agregado familiar é responsável por essa tarefa)

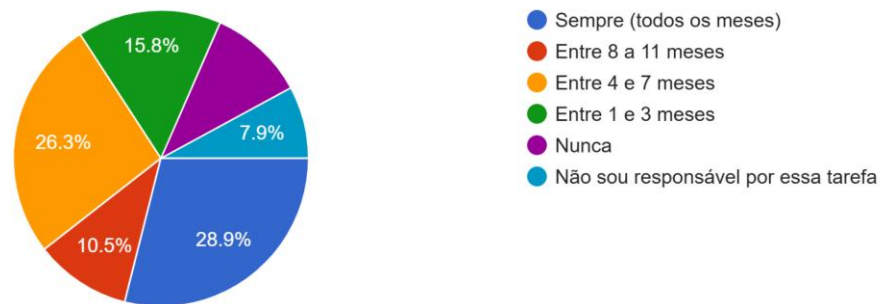
No último ano, e de forma aproximada, com que frequência cumpriu o seu orçamento mensal?

38 responses



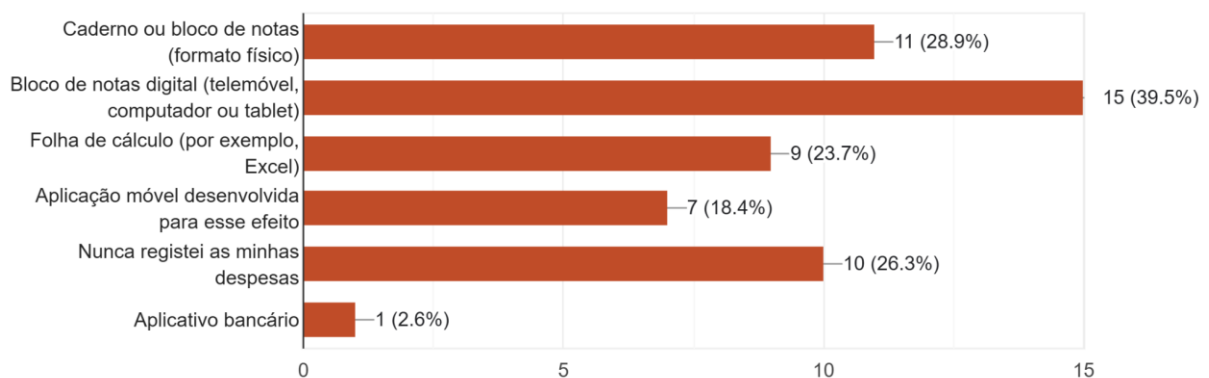
No último ano, e de forma aproximada, com que frequência conseguiu poupar?

38 responses



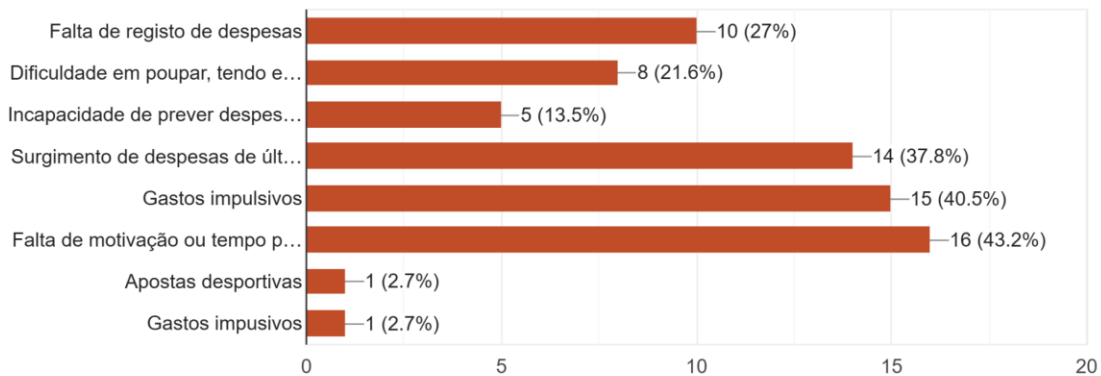
Quais dos seguintes tipos de registo de despesas já utilizou?

38 responses



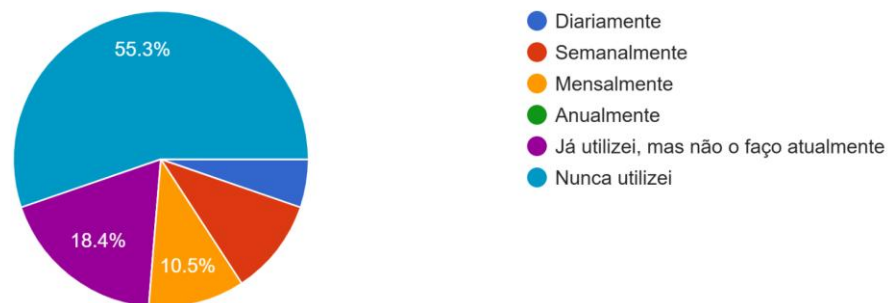
Dos seguintes desafios, quais considera serem os que mais o(a) impedem de controlar a sua gestão financeira?

37 responses



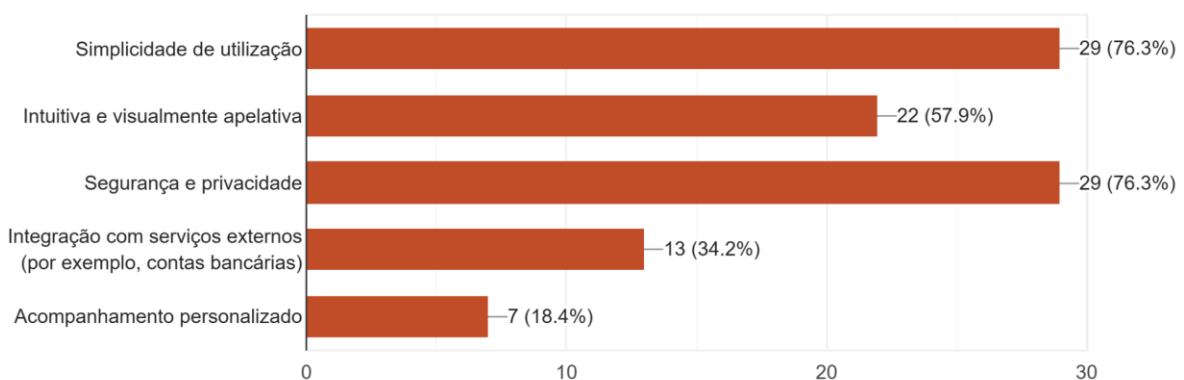
Com que frequência utiliza uma aplicação de gestão financeira?

38 responses



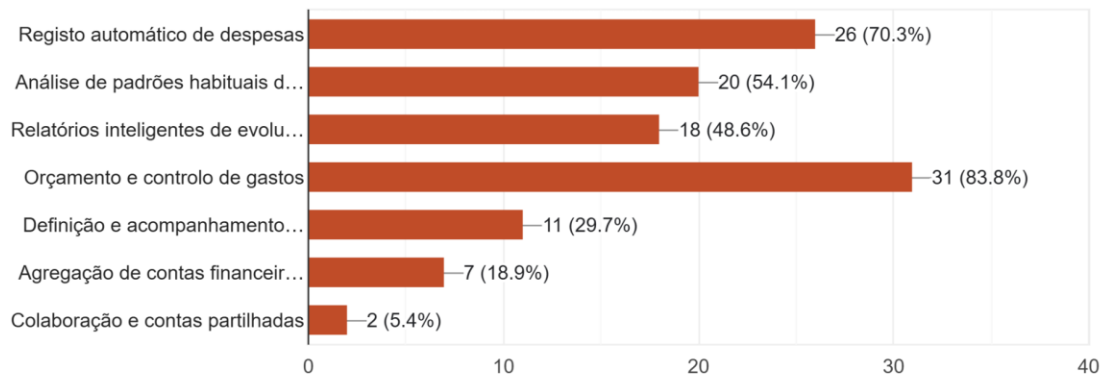
O que o(a) interessa neste tipo de aplicações? Selecione, por favor, até 3 interesses que considera mais relevantes.

38 responses



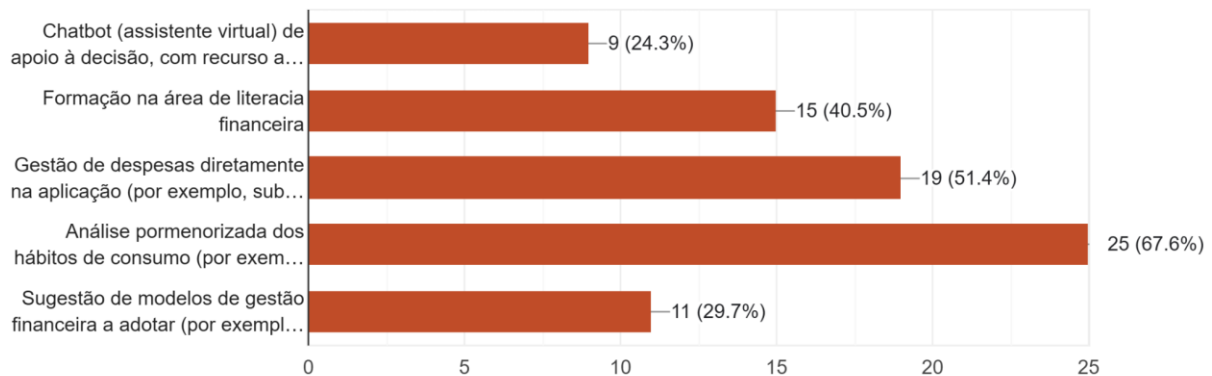
Quais das seguintes funcionalidades considera mais interessantes nas aplicações de gestão financeira que existem hoje em dia?

37 responses



Quais das seguintes inovações considera interessantes numa potencial nova aplicação de gestão financeira?

37 responses



Tem mais alguma sugestão de funcionalidade inovadora que gostaria que uma aplicação deste género incluísse?

4 responses

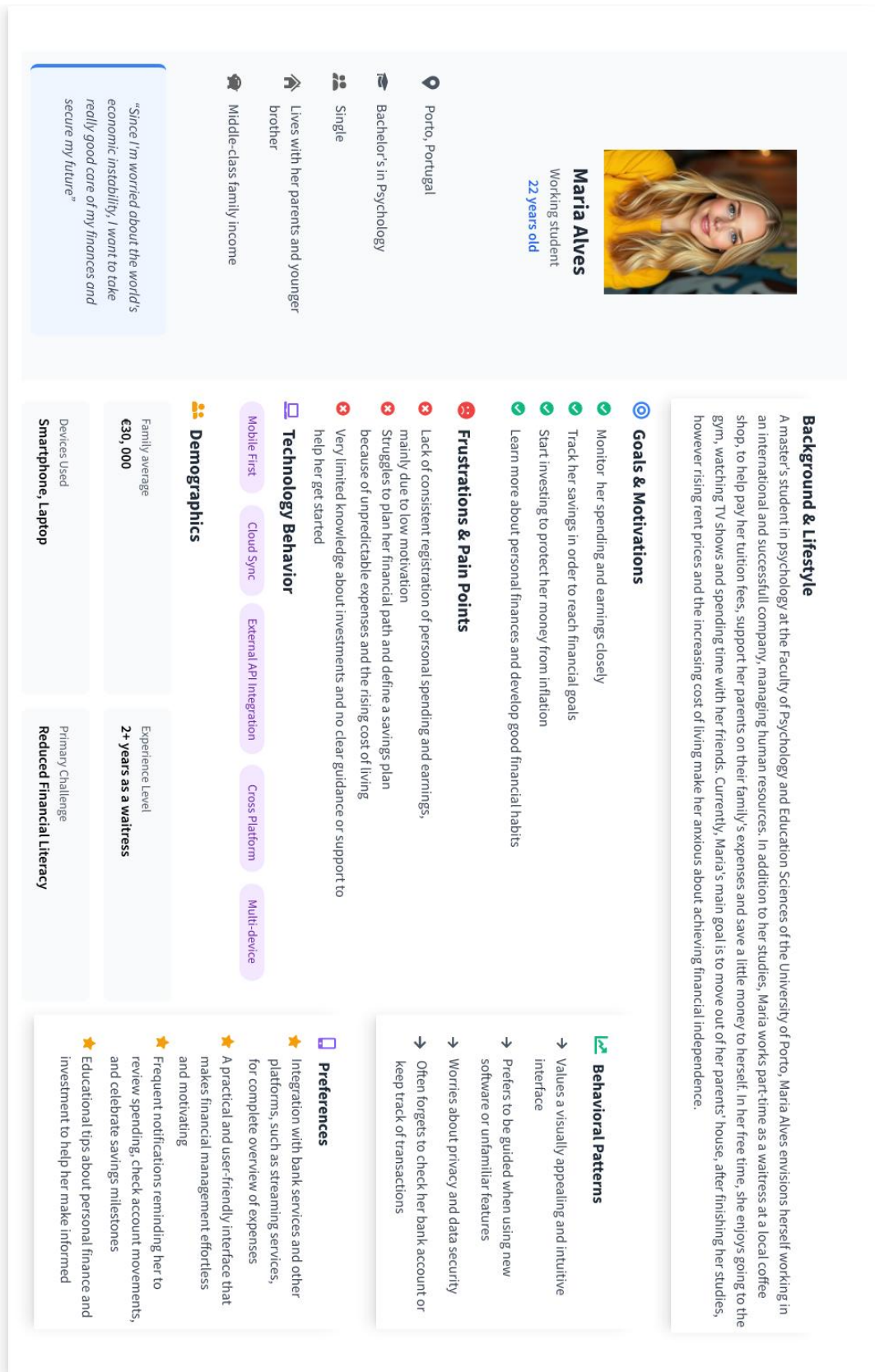
Não.

Painel de tendências (similar ao site investidor10 ou AUVF)

Alerta de preço de determinado ativo + notificação para tal

Annex III: Personas – visual representation

Persona 1: Maria Alves



Persona 2: Bernardo Martins



Bernardo Martins

Real Estate Agent

41 years old



Setúbal, Portugal



Real Estate Agent at Prime



Bachelor's in Business Administration



Married, in a long-term relationship



Lives with his wife and teenage son, and occasionally supports his parents



Medium to upper-middle income, commission-based

Goals & Motivations

✔

Monitor commission earnings from multiple property sales

✔

Prepare for his son's university education and his own retirement

✔

Generate monthly financial reports for tax purposes

✔

Gain better understanding of investment opportunities

Frustrations & Pain Points

✖

Difficulty reconciling payments from different clients

✖

Manual spreadsheet tracking is time-consuming and error-prone

✖

Limited investment knowledge, relies on intuition or friends' advice

✖

Struggles to plan long-term budgets due to unpredictable earnings

Technology Behavior

Mobile First

Cloud Sync

Bank Integration

Report Export

Multi-device

Demographics

Annual Income

€37,000 - €80,000

Experience Level

15+ years in real estate

Devices Used

Smartphone, Tablet, Laptop

Primary Challenge

Financial organization

Behavioral Patterns

→ Often forgets to check financial status

→ Prefers visual data representation over text

→ Needs quick access during client meetings

→ Values data security and privacy

Preferences

★ Simple, intuitive interface with minimal clicks

★ Mobile notifications for payment milestones

★ Integration with banking apps and services

★ Exportable reports for accountant collaboration

★ Interested in educational tips about saving and investment

"My income isn't always predictable — I just need a simple way to understand my spending and build savings for the future."

Background & Lifestyle

A seasoned agent for a well-known Setúbal real estate agency, Bernardo excels in his sales, but grapples with the inherent volatility of a commission-based income. This variability constantly complicates his attempts at financial planning. He keeps one eye on the market and global events, closely following the news and winding down with sports. However, when it comes to his own money, Bernardo relies more on instinct than spreadsheets. Despite efforts to manually log his commissions and track expenses in paper or Excel, his financial decisions—including small, friendly-advised investments—are often driven by the moment rather than a strategic overview.

HCI 2025/26 – MoneyBrew – Final report

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Annex IV: Heuristic evaluation report sent to group 03 (Fundly)

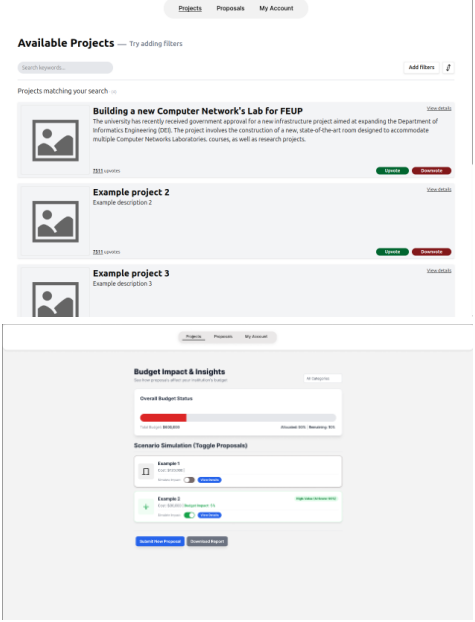
HCI Winter Semester 2025 - 2026

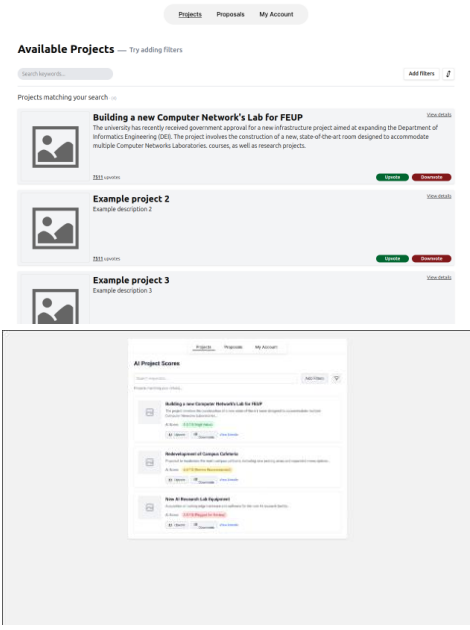
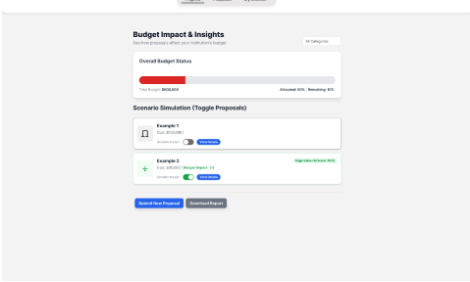
Heuristic Evaluation Report

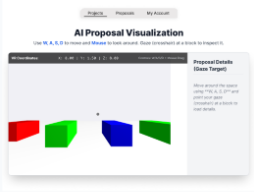
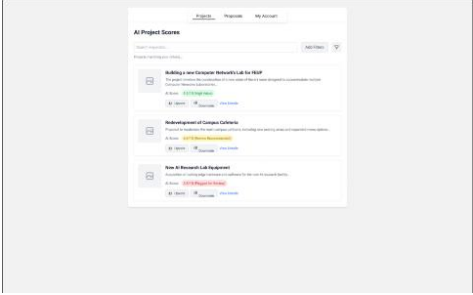
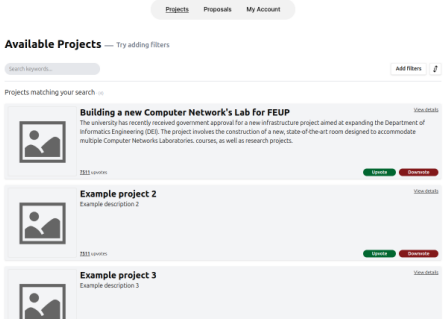
Class Nr.: LEIC03 - 12/11/2025 - Ana Sofia Teixeira

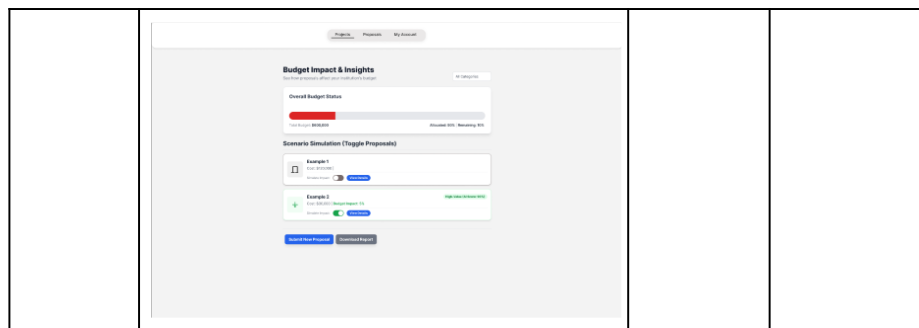
Group evaluated: 03 - Fundly

By group: 02 - MoneyBrew

Problem #	Issue (include screenshot)	Heuristic(s)	Severity (1-4)
1	The format of similar elements between pages, such as the top bar or the layout is not the same, making it ambiguous to the user about the similarity functionality of those elements 	4	2
2	The text "Try adding filters" is adding unnecessary content to the page due to the existence of the "Add filters" button close to it.	8	2

5	The necessity of these two separate pages seems unnecessary (adds needless complexity for the user) 	8	4
6	Connection between different pages is not clear (how to go from one page to the other) 	3	4

			
7	Lack of colored upvote and downvote buttons that exists in other pages 	4	3
8	Buttons with similar functionality and related to the same object, “submit new proposal” and “create new proposal”, have different colors 	4	2



Annex V: Heuristic evaluation report sent to group 04 (ZenBudget)

HCI Winter Semester 2025 - 2026

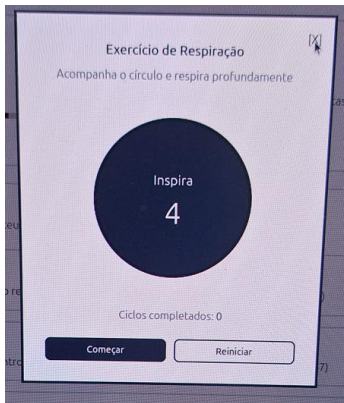
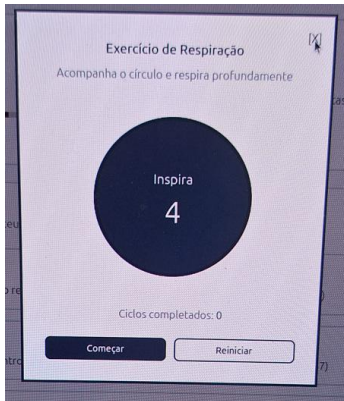
Heuristic Evaluation Report

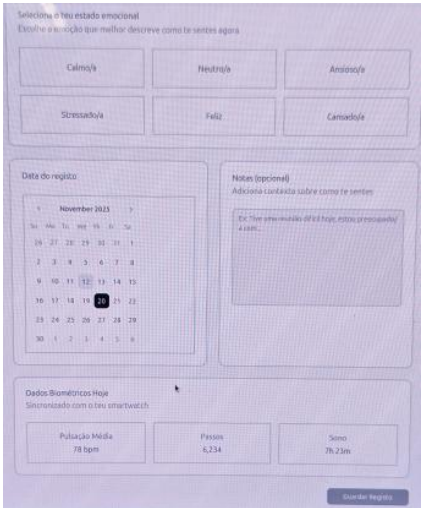
Class Nr.: LEIC03 - 12/11/2025 - Ana Sofia Teixeira

Group evaluated: 04 - ZenBudget

By group: 02 - MoneyBrew

Problem #	Issue (include screenshot)	Heuristic(s)	Severity (1-4)
1	We consider that the different breathing stages, “breathe in”, “pause”, and “breathe out”, should have different colors, to make it easier and more intuitive to distinguish the different steps of the cycle (although we understand that this did not necessarily need to be implemented at the lo-fi prototype phase). 	1	1

2	Exiting the breathing exercise window could be a clearer task, as it was not immediately evident for us, as users, when the exercise was ending nor how we could end it. Perhaps it would be better to have a third button next to “Pause” and “Reset”. 	3	2
3	Closing the breathing exercise window did not prompt a confirmation pop-up, to ensure the user definitely wants to quit the exercise. 	5	3

4	When logging an “emotional status” entry, the app currently allows the user to enter information for a date that is yet to pass, or, at the very least, does not prompt a pop-up alerting the user of that error. 	5	4
5	The side panel section in which the users can collect rewards is not evident, as "Community" is more suggestive of the place where users can exchange messages or see each others' profiles, and not so much a place to complete individual tasks, like collecting rewards. 	4	3

Annex VI: Heuristic evaluation report received from group 01 (TrustLend)

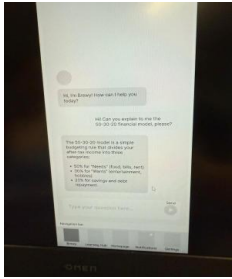
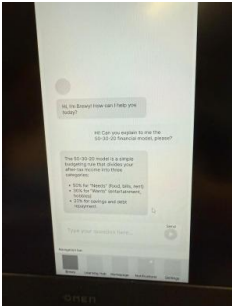
HCI Winter Semester 2025 - 2026

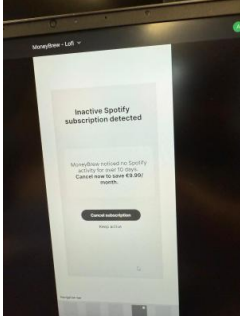
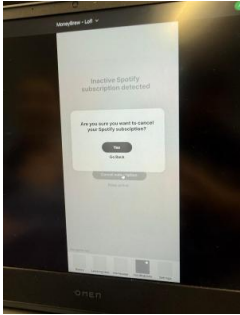
Heuristic Evaluation Report

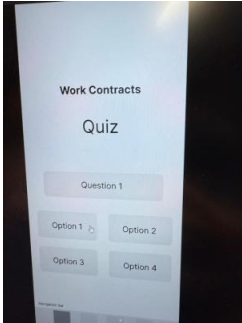
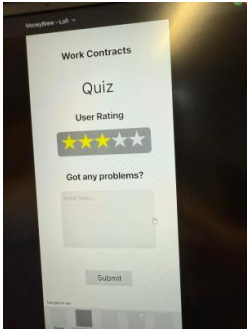
Class Nr.: LEIC03 - 12/11/2025 - Ana Sofia Teixeira

Group evaluated: 02 - MoneyBrew

By group: 01

Problem #	Issue (include screenshot)	Heuristic(s)	Severity (1-4)
1	Task 1 - The user should be able to see the date and time they sent their messages, and also be able to edit them in case of mistakes/needling to resend with more context. 	3 - User control and freedom	2
2	Task 1 - It would be good for the user to be able to send voice messages and images to obtain better guidance. 	7 - Flexibility and efficiency of use	2

3	Task 2 - The two options presented do not have the same weight in terms of contrast, pushing the user into making a specific decision. 	3 - User control and freedom	3
4	Task 2 - The user expects his choice to be “Yes” or “No”, but the given answers are “Yes” (a response) and “Go Back” (a system action). 	4 - Consistency and standards	2

5	Task 3 - The user has no way to know how many questions they have left. 	1 - Visibility of system status	2
6	Task 3 - The user cannot cancel or “dodge” having to give the quiz a rating at the end. They are forced to interact and submit a response, which can be very frustrating. 	3 - User control and freedom	3

Annex VII: Heuristic evaluation report received from group 06 (Growplify)

HCI Winter Semester 2025 - 2026

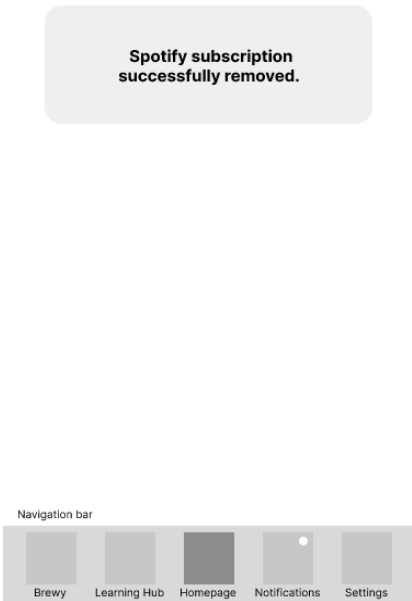
Heuristic Evaluation Report

Class Nr.: LEIC03 - 12/11/2025 - Ana Sofia Gomes Carrega e

Resende Teixeira

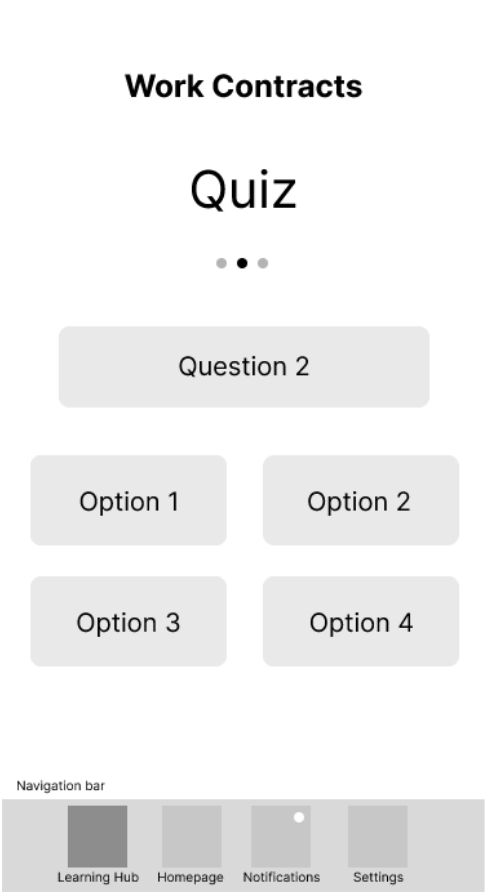
Group evaluated: 02 - MoneyBrew

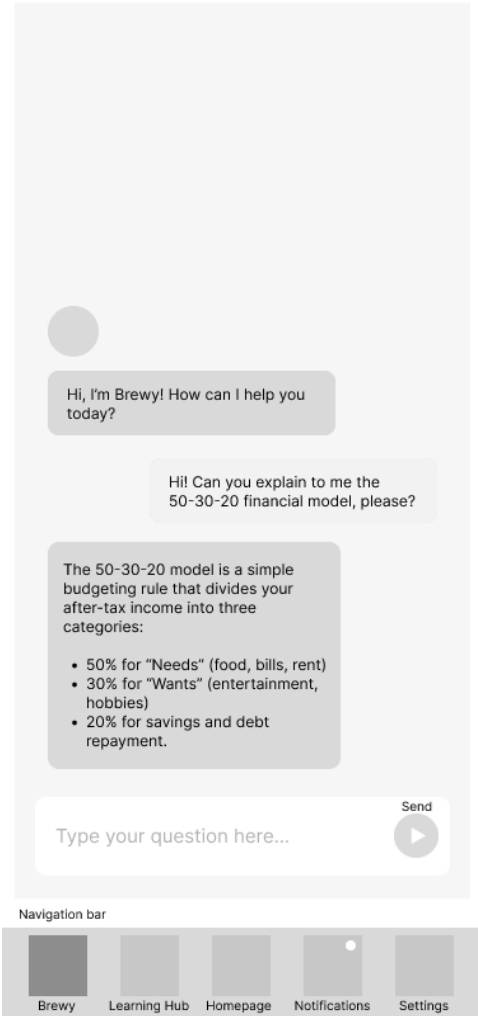
By group: 06

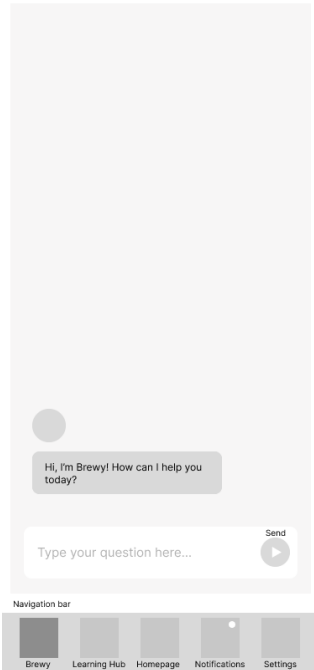
Problem #	Issue (include screenshot)	Heuristic(s)	Severity (1-4)
1	No option to reverse subscription cancellation, after confirming cancellation.  The screenshot shows a light gray rounded rectangle with the text 'Spotify subscription successfully removed.' in bold. Below it is a navigation bar with five items: 'Brewy', 'Learning Hub', 'Homepage' (which is highlighted with a dark gray background), 'Notifications', and 'Settings'.	3	2

2	After confirming the cancellation, it's not clear when the change takes effect or whether the user will keep access until the end of the billing period. (The image illustrates that, after cancellation, the system goes straight to the homepage, not making the system status clear to the user) Your financials at a glance Recent activity <table><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td>Transaction description</td><td>Amount</td></tr></table> Investments <table><tr><td></td><td>Balance</td></tr><tr><td>Investments graph</td><td>Latest updates</td></tr></table> Shopping list <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td>Items</td><td>Qnts.</td><td>Prices</td></tr></table> Navigation bar <table><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Brewy</td><td>Learning Hub</td><td>Homepage</td><td>Notifications</td><td>Settings</td></tr></table>							Transaction description	Amount		Balance	Investments graph	Latest updates																Items	Qnts.	Prices						Brewy	Learning Hub	Homepage	Notifications	Settings	1	1
Transaction description	Amount																																										
	Balance																																										
Investments graph	Latest updates																																										
Items	Qnts.	Prices																																									
Brewy	Learning Hub	Homepage	Notifications	Settings																																							

3	The quiz results screen only shows if it is correct or wrong without context. Users may not remember what each question was or which answer they gave. Work Contracts Quiz Results Time Spent xxm:yy Results Question 1 ✓ Question 2 ✗ Question 3 ✓ Restart Go Back Navigation bar Brewy Learning Hub Homepage Notifications Settings	6	3
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4	Once the quiz begins, there's no visible way to exit, pause it, or go back and forward between questions.  Work Contracts Quiz ... Question 2 Option 1 Option 2 Option 3 Option 4 Navigation bar Learning Hub Homepage Notifications Settings	3	2
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5	There's no visible help icon or information on what the chatbot can do and how it works.  Hi, I'm Brewy! How can I help you today? Hi! Can you explain to me the 50-30-20 financial model, please? The 50-30-20 model is a simple budgeting rule that divides your after-tax income into three categories: • 50% for "Needs" (food, bills, rent) • 30% for "Wants" (entertainment, hobbies) • 20% for savings and debt repayment. Type your question here... Send Navigation bar Brewy Learning Hub Homepage Notifications Settings	10	1
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6	The chatbot's introduction message sounds too generic, lacking clear context about its financial focus. It should be made clear to the user what is the scope of the chatbot, instead of relying on the user knowing/remembering. 	6	3
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